

Crusher to the Bridge:
Multi-Pathogen Outbreak Simulation and
Intervention Analysis Across Maritime Vessel Classes

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Abstract

Crusher to the Bridge (CTB) is an individual-based model for simulating infectious-disease transmission, detection, and response on cruise ships. Drawing on 77,310 simulation runs across four vessel classes and ten pathogen profiles (69,750 original campaign runs, 1,200 synthetic-recovery runs, and 6,360 VSP degradation runs), this report develops two central findings. First, publicly reported cruise-ship norovirus attack rates are intervention-shadowed observations: with no response, the calibrated norovirus parameterisation produces platform-dependent attack rates of 10.6–23.4%, but syndromic detection and VSP-like escalation compress these to a 4.6–7.1% band that matches CDC Vessel Sanitation Program surveillance records. This 10.6–23.4% gradient is a model output under the calibrated parameterization; no empirical uncontrolled cruise attack rates exist to constrain it directly, and the identifiability experiment in Section 5.9 shows that alternative parameterizations along the dose–alpha ridge could reproduce the same post-response band with a different uncontrolled gradient. Second, the return on mid-voyage surveillance improvements is small once any syndromic response is active (generally less than one percentage point of absolute attack-rate reduction), whereas pre-boarding wearable screening can deliver value-of-information exceeding \$10,000 per passenger on expedition-class vessels under the mid-cost scenario, ranging from \$2,500 under conservative assumptions to \$63,000 under high-cost assumptions. A multi-pathogen pre-boarding decision model, grounded in k-indexed outbreak surfaces from 17,600 dedicated simulation runs, shows that screening value varies sharply by pathogen: SARS-CoV-2 and measles offer the widest prevalence windows for cost-effective interception, while influenza saturates too quickly for screening to help on large ships. The model, codebase, and campaign data are publicly available at https://github.com/bckirkup/Crusher_to_the_Bridge.

Keywords: agent-based model; maritime epidemiology; norovirus; SARS-CoV-2; cruise ship; surveillance; pre-boarding screening; CONTAM; outbreak response; value of information

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Chapter 1

Introduction

1.1 Scientific claim

Cruise ships are useful epidemiologic instruments because they combine dense occupation, repeated social schedules, medical ascertainment, and standardized regulatory reporting in a setting where introductions, transmission, and response can be observed on a voyage time scale. They are also operationally important: a failure of onboard control can affect passengers, crew, ports, and shore-side health systems before the voyage is recognized as an outbreak. The central question of this monograph is whether physical differences among vessel classes remain visible after standardized shipboard response, or whether the response system compresses observed attack rates enough to hide the underlying vessel gradient.

1.2 Why cruise ships matter

Commercial cruising is a large population movement, not a marginal travel niche. Before the COVID-19 disruption, industry reports expected roughly 30 million passengers to cruise in 2019, and CLIA's 2025 report recorded 34.6 million ocean-going cruisers in 2024 with 37.7 million forecast for 2025 [1, 2]. Those passengers move through ports, airports, hotels, excursions, and destination communities. The epidemiologic unit is therefore larger than the hull: a ship concentrates exposure during the voyage, then redistributes infected or exposed travelers across jurisdictions.

Ships matter scientifically because they occupy a middle scale between households and cities. Cabins create household-like clusters; restaurants, buffets, theaters, bars, spas, tenders, elevators, and corridors create repeated shared venues; crew work networks bridge passenger-facing and crew-only spaces; and the population is present long enough for respiratory and enteric pathogens to pass through multiple generations [3, 4]. The contact graph is more stable than a city contact graph because cabins, dining assignments, work duties, and daily schedules recur. It is less controlled than a hospital ward because passengers choose activities, report symptoms voluntarily, and leave

the ship at the end of the voyage. This intermediate scale makes cruise ships a useful test bed for models that must combine individual movement, environmental reservoirs, shared air, clinical observation, and reactive protocols.

Ships also matter operationally because they are regulated and instrumented. Covered vessels entering the United States from foreign ports report acute gastroenteritis (AGE) through the CDC Vessel Sanitation Program (VSP), keep medical logs, undergo unannounced sanitation inspections, and interact with port-health and quarantine authorities [5, 6]. International Health Regulations (IHR) obligations, WHO ship-sanitation guidance, European SHIPSAN activity, national biosecurity systems, and industry protocols add further layers of reporting and control [4, 7]. As a result, a cruise outbreak is rarely an unobserved epidemic curve. It is an epidemic curve filtered through care seeking, case definitions, reporting thresholds, sanitation procedures, isolation capacity, port decisions, and public communication.

1.3 A ship is both a transmission setting and an observation system

The same properties that make ships useful can bias their data. VSP surveillance began in 1975 after repeated shipboard gastrointestinal outbreaks and has maintained diarrheal-illness records since April 1 of that year [5, 8]. Its modern AGE ladder requires routine arrival reports, a special report and active monitoring at 2% cumulative AGE in passengers or crew, and outbreak classification at 3% or more in either population [5, 6]. The 3% threshold is an administrative and operational rule. It selects public records, directs CDC attention, and usually intensifies onboard response. It is not a biological threshold at which norovirus or another pathogen changes its transmission law.

This dual use of the threshold is central. A VSP record is produced after passengers or crew develop symptoms, decide or are required to report to the medical center, meet a syndromic case definition, enter the cumulative numerator of the reported AGE count, and trigger monitoring or outbreak actions. Public reports identify ships, voyage dates, denominators, reported AGE counts, symptoms, likely etiologies when confirmed, and response measures, but they omit unreported illness, sub-threshold voyages, some cases outside the arrival window, and events outside VSP jurisdiction [5, 9]. During 2006–2019, MIDRS received 37,258 voyage reports from 252 ships, but CDC characterized the incidence estimates as underestimates because only qualifying medical-center presentations were counted [5]. Thus the public VSP dataset is not a census of onboard infection. It is a threshold-selected, clinic-ascertained, jurisdictionally bounded record of syndromic illness.

The implication for simulation is direct. A model calibrated to VSP records as if they were uncontrolled final attack rates will misread the object being fitted. The empirical target includes transmission, reporting, and response. This monograph therefore treats VSP as an observation-and-intervention system, not as a passive data feed. The phrase “VSP shadow” used later refers to

this coupling between what is measured and what is done once the measurement crosses a response boundary.

1.4 Historical outbreak record

Norovirus and other causes of AGE dominate the routine cruise-outbreak literature. Reviews and CDC summaries show that cruise norovirus is multi-route: person-to-person spread, cabin exposure, food-service exposure, shared surfaces, vomiting events, and incomplete environmental reset can all contribute [10–13]. The 2024 systematic review identified 45 outbreaks on 26 ships and estimated pooled attack rates of 7% among passengers and 2% among crew, with an ill cabin mate as the strongest recurring risk factor [10]. Older reports remain useful because they show persistence across voyages, mixed strains, and passenger behaviors that alter reporting and spread [14, 15]. Recent VSP investigations continue to show the same operating problem: by the time a threshold is crossed, line listing, isolation, enhanced cleaning, specimen collection, spatial investigation, and CDC consultation have already become part of the epidemic process [16].

Respiratory outbreaks broaden the question from surfaces and food-service networks to shared air, cabin confinement, crew mobility, and port coordination. Diamond Princess remains the most examined modern cruise outbreak. Passengers and crew experienced quarantine and cabin confinement during the early COVID-19 pandemic, so the observed 19.2% final positive fraction is a partially mitigated outcome rather than an unmitigated baseline. Extensive virologic, epidemiologic, and environmental data were collected [17–19]. The ship showed high epidemic potential in a confined population and also showed how late quarantine, shared cabins, and crew service needs can limit response. Route attribution remains disputed. Azimi and colleagues estimated an important aerosol component, whereas cabin-based analyses argued against substantial cross-cabin transmission through ventilation [20–22]. A useful ship model should therefore allow alternative respiratory-route assumptions rather than treating one route estimate as settled fact.

Other COVID-19 cruise events show that Diamond Princess was not an isolated anomaly. Reviews through October 2020 identified many affected ships and voyages, with Diamond Princess and Ruby Princess as major outliers and Greg Mortimer reporting attack rates above 50% [23–25]. The Ruby Princess inquiry is especially important operationally because it connects shipboard illness recognition with port decision-making and subsequent community transmission [26]. These events are not interchangeable with VSP AGE outbreaks, but they share a common structure: boundary decisions before boarding and at port can determine whether onboard surveillance ever has a reasonable chance to succeed.

The shipboard literature also includes influenza, measles and other vaccine-preventable diseases, Legionella, foodborne and waterborne outbreaks, wastewater surveillance, and emerging or rare events [27–32]. The MV *Hondius* Andes-hantavirus event illustrates a different edge case: a novel or unexpected pathogen can appear in a confined maritime setting even after COVID-19 preparedness

has made respiratory screening and isolation more familiar [33]. For modeling, these records justify a multi-pathogen framework. They do not justify assuming that one pathogen’s route weights, reporting probability, or response efficacy transfers to another.

1.5 The central question

The monograph asks whether physical differences between vessel classes remain visible after a standardized response, defined here as the same syndromic-reporting threshold logic, isolation and sanitation protocol family, and diagnostic-cascade structure applied across vessel classes while allowing platform-specific operating capacity parameters to remain explicit. The question is narrow but important. Vessels differ in population size, cabin density, deck topology, dining patterns, crew fraction, medical capacity, ventilation architecture, and port itinerary. Those differences can affect the probability that an introduction becomes an outbreak and the conditional final size if it does. Yet public records are generated after response, and response can compress final attack rates. A flat observed distribution across vessels could mean that ship architecture is epidemiologically irrelevant. It could also mean that a stronger response truncates outbreaks before physical differences fully express themselves.

The distinction changes both science and policy. For science, the relevant target is not a single final attack rate. It is the joint distribution of introduction, stochastic fade-out, threshold triggering, time to detection, conditional attack rate, and response burden. For operations, the relevant decision is not whether a detector lowers attack rate in a perfectly observed epidemic. It is whether the detector improves decisions under real boundary conditions: infected travelers may board, symptoms may be hidden, reporting may lag, and port authorities may have to act before laboratory certainty.

Pre-boarding screening is therefore introduced here as a boundary problem. Pre-boarding tests, questionnaires, vaccination documentation, symptom checks, and travel-history rules act before the ship’s internal observation system has data. They can reduce introductions, but imperfect sensitivity and passenger incentives mean they also create false reassurance. Mid-voyage surveillance faces the opposite problem: it observes the onboard epidemic after some transmission has already occurred. The comparison between boundary control and mid-voyage detection recurs in the intervention chapters because it determines whether advanced surveillance adds information early enough to change outcomes.

1.6 Approach and data used in the monograph

The empirical and computational base is the open-source *Crusher to the Bridge* model, available at https://github.com/bckirkup/Crusher_to_the_Bridge. The monograph uses 77,310 simulation runs across 4 pathogens and 4 vessel platforms, with 285 CDC VSP records used as the

empirical reference set for AGE calibration. The model is treated as a coupled system of infection biology, ship movement, environmental transport, observation, and protocol response. That framing follows directly from the cruise evidence: onboard transmission and onboard measurement cannot be separated once thresholds activate control measures.

The four focal pathogens represent different operating regimes. Norovirus anchors the VSP AGE problem and tests food, fomite, cabin, and reporting dynamics. SARS-CoV-2 anchors respiratory-route uncertainty and the Diamond Princess literature. Influenza provides a common respiratory pathogen with documented cruise outbreaks and shorter generation times. Measles tests the boundary-introduction problem for a highly transmissible pathogen whose onboard risk depends on susceptible pockets and vaccination documentation. Other pathogens in the literature, including Legionella, foodborne bacteria, wastewater-detected viruses, and the MV *Hondius* hantavirus event, are used to interpret scope and limits rather than to claim a single universal shipboard transmission law.

1.7 Roadmap

Chapter 2 describes the CTB architecture: agent states, ship platforms, route-weighted dose accounting, environmental transport, observation instruments, and protocol logic. It positions the model against prior confined-environment and ship-specific models and explains why a cruise model must couple movement, infection, instrumentation, and response rather than simulate transmission alone.

The calibration chapters then address measurement. They separate latent infection from reported illness, show why VSP records are post-response observations, and estimate pathogen and platform behavior against the relevant surveillance endpoint rather than against an imagined uncontrolled curve. The VSP chapter develops the 3% threshold as both an observation rule and an intervention trigger and uses the 285 public records as a threshold-selected AGE reference set.

The intervention chapters evaluate what ship operators and public-health authorities can change. They compare syndromic response, wastewater, wearables, multiplex testing, contact tracing, isolation, and pre-boarding controls across pathogens and vessel classes. The recurring decision question is whether an instrument changes action early enough to matter, not whether it detects infection in principle.

The concluding synthesis returns to the central question. If vessel-class gradients persist after standardized response, then ship design and operating patterns remain visible in observed outbreaks. If response compresses those gradients, then public records can look deceptively similar across vessels while the uncontrolled risk remains heterogeneous. The monograph’s contribution is to make that distinction explicit, quantify it across simulation campaigns, and provide a reproducible framework for testing shipboard outbreak decisions.

Chapter 2

Model Architecture

2.1 Prior confined-environment ABMs

CTB is designed to model a regulated, instrumented confined environment rather than an unobserved epidemic curve. Prior confined-environment models provide parts of this capability, but they usually emphasize one layer at a time: individual contacts, spatial movement, route attribution, air-flow, environmental persistence, or a small intervention set [20, 34–42]. The distinction matters for cruise ships because the empirical endpoint is already altered by case recognition, isolation, cleaning, food-service changes, and regulatory reporting [5, 10, 16, 24]. CTB therefore treats infection generation, observation, and response as one coupled system.

Table 2.1 summarizes the comparison used to position the model. The table includes agent-based models and closely related mechanistic ship or indoor-air models because cruise-ship transmission lies at the intersection of individual mobility, repeated contacts, environmental reservoirs, and multizone environmental transport [3, 20, 41, 43]. Kerkmann 2024 and Kou 2021 were not cited because the literature audit could not verify those items as requested. The defensible novelty claim is narrow: among the accessible comparators reviewed here, this combination of individual route-weighted dose response, CONTAM-style multizone HVAC coupling, multiple pathogens, VSP-style escalation, wearable biosurveillance, and operational decision logic was not found in a single cruise-ship framework. This is a scoped comparison, not a systematic-review claim.

Table 2.1: Prior confined-environment transmission models and their relationship to CTB.

Study	Environment and type	Transmission representation	Calibration or validation	Limitation relative to CTB
Hunter et al. [34]	Open-data ABM for Irish towns	NetLogo spatial SEIR with co-location contacts, census and land-use driven agents, and stochastic disease periods	Tested against the 2012 Schull measles outbreak and transferred to multiple towns	No ship geometry, HVAC transport, route-specific dose accumulation, or regulatory response layer
Pan et al. [44]	Reported cruise-ship ABM for healthy ship design	Spatial and operational interventions are indicated by metadata, but detailed mechanics were not available in the reviewed text	Not recoverable from accessible full text in the literature review	Potentially adjacent to CTB, but feature-level comparison could not be verified
Rankin [35]	NetLogo3D Diamond Princess ABM	Stochastic SEIR agents with passenger and crew movement, proximity contacts, and quarantine-restricted movement	Compared with the observed Diamond Princess attack fraction and a previous SEIR estimate	No explicit aerosol concentration, fomite route, dose-response, or dynamic instrument-driven response
Stephenson et al. [37]	Acute-care ABM for <i>Clostridioides difficile</i>	Patients, healthcare workers, contaminated rooms, antibiotic effects, hand/contact compliance, cleaning, stewardship, and vaccination scenarios	Parameters assembled from clinical and environmental evidence	Hospital-specific organism and intervention stack; no ship geometry, HVAC coupling, or voyage response logic
Barker et al. [38]	Adult acute-care hospital ABM for <i>C. difficile</i>	Patients, nurses, physicians, visitors, environmental contacts, hand hygiene, cleaning, screening/isolation, and stewardship	Disease-state transitions calibrated from literature and hospital data	Generic hospital layout; intervention rankings may not transfer to particular vessels or pathogens

Study	Environment and type	Transmission representation	Calibration or validation	Limitation relative to CTB
Scaria et al. [39]	Hospital-specific <i>C. difficile</i> ABM	Facility layout, ward behaviors, environmental works, and colonization pressure	Validated against 2013–2018 observations from a 426-bed hospital	Strong facility validation, but no cruise operations, passenger turnover, or respiratory/enteric multi-pathogen abstraction
Almagor et al. [40]	Hospital for antibiotic-resistant bacteria ABM	Patient–healthcare-worker adjacent-patient contacts, antibiotic-susceptibility, and antibiotic stewardship	Literature-informed typical hospital	Generic hospital setting; no explicit environmental airflow, dose-response, or maritime observation system
Azimi et al. [20]	Mechanistic Diamond Princess pathway model, not a conventional ABM	Markov-chain environmental states separating short-range, fomite, aerosol, and large-droplet pathways	Large scenario sweep accepted by epidemic-curve fit criteria	Aggregated population states rather than autonomous ship agents and operational protocols
Xia et al. [41]	CONTAM plus Wells–Riley risk on a passenger ship	Multizone airflow and airborne-virus risk under prescribed occupant assumptions	Applied to an operating-vessel case without outbreak-curve validation in the retrieved evidence	Strong air-transport layer, but not an individual outbreak model with clinical observation and SOP response
D’Orazio et al. [36]	Public-building ABM calibrated using Diamond Princess	Occupant movement, close-contact duration, masks, density reduction, and infectious-entry restrictions	Transmission coefficient tuned to the Diamond Princess epidemic segment and used for building scenarios	No ship operating model, multizone HVAC solver, or multi-pathogen dose framework

Study	Environment and type	Transmission representation	Calibration or validation	Limitation relative to CTB
Zheng et al. [42]	Cruise-ship individual contact network and SEIR-like model	Cabins, shared spaces, change rate, UVGI, quarantine	Validated against a previous cruise-ship outbreak	Ship-specific intervention comparison without the surveillance cascade, multiple pathogens, or CTB’s route-weighted dose ledger

This comparison motivates the architectural choice. The model must represent where people are, what route-specific dose they receive, which instrument would observe the event, and what response would be triggered. A pure contact ABM cannot answer airflow questions. A pure CONTAM or Wells–Riley calculation cannot answer quarantine-compliance or diagnostic-delay questions. A static intervention sweep cannot represent an escalation system in which the intervention exists only after the ship generates sufficient evidence.

2.2 CTB architecture overview

CTB is implemented in the public repository https://github.com/bckirkup/Crusher_to_the_Bridge and documented in code-grounded DeepWiki pages for the architecture overview, transmission core, and reactive protocol engine.¹ The codebase uses a decoupled architecture: a central orchestrator owns the epoch loop and passes state among engines for mobility, infection biology, transport, observation, protocols, cost, and telemetry. The design is intentionally modular because shipboard outbreaks couple processes with different natural units: individuals, zones, airflows, samples, SOPs, and resource ledgers. This follows the general ABM practice of representing behavior and intervention states explicitly while keeping mechanistic submodels testable against their own data streams [45–49].

Each simulation is a discrete-time voyage. The present implementation uses hourly epochs, matching the transport bridge constant `HOURS_PER_EPOCH = 1.0` and the per-epoch orchestrator design. Let $t = 0, 1, \dots, T$ index epochs. At each epoch, CTB advances the following state vector:

$$\mathcal{S}(t) = \{X_i(t), L_i(t), B_i(t), C_i(t), M_{z,p}(t), F_{z,p}(t), A_{z,p}(t), R(t)\}, \quad (2.1)$$

¹DeepWiki pages used here are the indexed https://deepwiki.com/bckirkup/Crusher_to_the_Bridge/1-architecture-overview, https://deepwiki.com/bckirkup/Crusher_to_the_Bridge/1.2-four-pathway-transmission-core, and https://deepwiki.com/bckirkup/Crusher_to_the_Bridge/1.6-reactive-protocol-engine.

where X_i is the disease state of agent i , L_i is location, B_i is behavioral or compliance state, C_i is cabin or class assignment, $M_{z,p}$ is airborne pathogen mass for pathogen p in zone z , $F_{z,p}$ is surface or food-associated mass, $A_{z,p}$ is the set of instrument observations available for zone or agent targets, and $R(t)$ is the regulatory or SOP response state.

The epoch loop is ordered to avoid circular interpretation of a response. First, the Korkin-derived ship engine updates agent movement and disease progression. Second, route-specific exposure is calculated from current locations and the environmental pools carried over from the previous update. Third, the transmission core samples infection from the accumulated dose and records exposure matrices. Fourth, aerosol, surface, food, and environmental pools are updated for subsequent epochs, either natively or through the CONTAM branch. Fifth, observation instruments sample agents, surfaces, air, wastewater, wearable streams, or clinical specimens. Sixth, stoplight logic converts raw instrument results into GREEN, AMBER, or RED signals. Seventh, the protocol engine evaluates SOP triggers, applies modifiers to the next epoch, and debits activation or recurring costs. Finally, telemetry is serialized as simulation history, artificial-lab-notebook records, contact tracing matrices, and resource ledgers.

Agents are initialized by `orchestrator_init.py` from the platform spatial layout, role fractions, pathogen profiles, and voyage configuration. The initialization layer assigns passenger or crew roles, home zones, cabin-mate sets, graywater collection zones, isolation-unit capacity, and pathogen seeding. Cabin grouping is explicit: agents assigned to a `Cabin_Corridor` are grouped into stateroom-size cohorts, and those cabin-mate IDs are later used to distinguish household-like exposure from non-mate corridor exposure during confinement. This is a material distinction for ship simulations because quarantine does not remove a person from the ship air system and does not eliminate cabin-mate exposure.

The spatial model is a graph of zones rather than a continuous geometry. Zones carry attributes such as type, traffic class, deck, volume, cabin ventilation type, cabin size, maximum occupancy, dining service type, and food-contamination multiplier. This representation is detailed enough to support cabin corridors, dining venues, galleys, medical spaces, public rooms, engineering spaces, graywater collection, and HVAC nodes while still allowing large campaign sweeps. The monograph campaigns are a 69,750-simulation-run programme across 12 named campaigns and the Starfleet stress test (Appendix A.4), covering four calibrated pathogens plus six stress-test profiles and four vessel platforms, with 285 CDC VSP empirical outbreak records used as the calibration reference set. Those records are interpreted as threshold-selected, clinic-ascertained AGE observations rather than as a census of all onboard infection [5, 6, 8, 50].

2.3 Agent state and pathogen seeding

For pathogen p , agent i has a disease state

$$X_{i,p}(t) \in \{S, E, I_a, I_s, R, Q\}, \quad (2.2)$$

where S is susceptible, E exposed, I_a asymptomatic infectious, I_s symptomatic infectious, and R recovered or no longer infectious. Q is an operational confinement or isolation flag layered on top of the biological state rather than a mutually exclusive disease state. The implementation stores these quantities through the infection-dynamics bridge and pathogen-aware agent methods rather than a single global SIR label. Multiple concurrent pathogens are therefore represented as pathogen-specific infection histories plus separate observation and response flags.

Pathogen seeding can occur at initialization or during the voyage. The initialization layer loads active pathogen profiles and infects configured initial agents or introduces pathogen mass into environmental reservoirs. Mid-cruise introductions are handled as epoch events so that port visits, embarkation layers, or late infectious arrivals can be represented without changing the core transmission equations. Let I_p^0 be the set of initially infected agents for pathogen p and $J_p(t)$ be any introduction process during the voyage. Then the infected set evolves by

$$\mathcal{I}_p(t+1) = [\mathcal{I}_p(t) \setminus \mathcal{R}_p(t)] \cup \mathcal{N}_p(t) \cup J_p(t), \quad (2.3)$$

where $\mathcal{N}_p(t)$ is the set of new infections produced by the dose-response step and $\mathcal{R}_p(t)$ is the set leaving the infectious state through recovery, removal, or response-mediated isolation from the transmissible pool. Imported-versus-secondary provenance is preserved in the separate event and transmission ledger, not by the set recursion itself; that ledger is what supports offspring distributions and fade-out probability.

Shedding is profile-specific. The common implementation form is a log-scale shedding curve adjusted by a fitted dose parameter:

$$V_{i,p}(t) = 10^{s_p(\tau_{i,p}(t)) - d_{\text{adj},p}}, \quad (2.4)$$

where $s_p(\tau)$ is the pathogen-specific log shedding trajectory at time since infection or symptom onset τ , and $d_{\text{adj},p}$ maps biological shedding units to effective simulation dose. This equation is the interface between pathogenesis and transmission. It allows norovirus, SARS-CoV-2, measles, Legionella, or other configured pathogens to share the same route ledger while retaining different shedding timing, dose-response parameters, route weights, and environmental persistence. The separation is required because norovirus RNA shedding can persist long after acute illness, SARS-CoV-2 and influenza infectiousness peak near symptom onset, measles has a presymptomatic infectious window, and Legionella is primarily an environmental-source hazard [30, 51–56].

2.4 Four-pathway transmission core

The transmission engine is called `TransmissionCore` in `engines/transmission_core.py`. It began as a four-pathway engine with direct contact, short-range droplet, long-range airborne transport, and fomite deposition. The current implementation retains those four primary pathways and extends them with food contamination and environmental-source pathways for enteric and colonized-environment pathogens. For formal notation, define

$$\mathcal{R} = \{\text{contact, droplet, airborne, fomite, food, environmental}\}. \quad (2.5)$$

When a pathogen profile does not enable food or environmental-source transmission, the corresponding dose terms are zero. The route set is motivated by cruise evidence for cabin and direct spread, contaminated surfaces, food-service exposures, and respiratory shared-air transmission, while retaining uncertainty where route attribution is disputed. Some cited outbreak reports document multiple plausible routes rather than estimating route interactions, so the route weights are scenario parameters rather than empirical fractions [12, 20, 21, 43, 57, 58].

The route-weighted dose received by agent i for pathogen p in epoch t is

$$D_{i,p}^{\text{raw}}(t) = \sum_{r \in \{\text{contact, droplet, airborne, fomite, food, environmental}\}} D_{i,p,r}(t), \quad D_{i,p}^{\text{total}}(t) = s_i(t) \sum_r w_{p,r} D_{i,p,r}(t), \quad (2.6)$$

Here $w_{p,r}$ is the pathogen-specific transmission-route weight, $D_{i,p,r}$ is the raw route dose delivered by the pathway module, and $s_i(t)$ is the susceptibility multiplier applied after route weighting. The code resolves absent route weights to identity values, applies each route weight once in `_apply_route_weights`, and then applies susceptibility multipliers before infection sampling.

The implemented default infection probability uses the approximate beta-Poisson form specified in the pathogen profile and in `engines/transmission_core.py`. The exact Teunis confluent-hypergeometric equation is retained in Chapter 4 as a biological shape prior and sensitivity target, but the monograph campaign results reported here were generated by the approximate implementation below using the profile parameters in Appendix A.2 [59–61]:

$$P_{i,p}(\text{infection} \mid D) = 1 - \left(1 + \frac{D_{i,p}^{\text{total}}}{\beta_p}\right)^{-\alpha_p}. \quad (2.7)$$

An exponential dose-response option is also implemented for pathogens or calibration settings that use a single rate constant, which is useful when the available evidence supports only an effective risk parameter rather than a full host-heterogeneity distribution [60, 62]:

$$P_{i,p}(\text{infection} \mid D) = 1 - \exp(-\kappa_p D_{i,p}^{\text{total}}). \quad (2.8)$$

Only susceptible, non-immune agents are sampled. If infection occurs, the event is recorded with the

dominant pathway, total dose, pathogen ID, zone, and pathway breakdown. Offspring summaries in this work use a proportional exposure-attribution rule: a new infection is allocated across source agents in proportion to their contribution to the target’s route-specific dose within the infection epoch; if only aggregate environmental mass is available, the event is retained for route and zone attribution but not assigned to a unique person-to-person parent.

2.4.1 Galley and crew segmentation

Food contamination creates a non-linear amplification pathway for enteric pathogens because a single back-of-house (BOH) food handler can expose an entire dining cohort during one meal epoch. In the current implementation, BOH crew handle food items that are served to passengers in dining venues, and an infected BOH worker contaminates the venue food pool at a rate proportional to their shedding level. The food pool accumulates by venue during food-service epochs and decays between meals, so repeated service periods can generate venue-specific exposure histories rather than a single shipwide food dose.

For dining venue v , the food-associated reservoir for pathogen p can be written as

$$F_{v,p}(t+1) = (1 - \delta_{p,\text{food}})F_{v,p}(t) + \sum_{j \in \mathcal{B}_v(t) \cap \mathcal{I}_p(t)} \xi_{p,\text{food}} V_{j,p}(t), \quad (2.9)$$

where $\mathcal{B}_v(t)$ is the set of BOH food handlers assigned to venue v during epoch t , $V_{j,p}(t)$ is the shedding intensity of infected worker j , $\xi_{p,\text{food}}$ converts shedding into food-pool contamination, and $\delta_{p,\text{food}}$ is between-meal decay or removal. Passengers and crew eating in that venue receive an additional route-weighted dose

$$D_{i,p,\text{food}}(t) = \omega_{iv}(t) F_{v,p}(t), \quad D_{i,p,\text{food}}^{\text{weighted}}(t) = w_{p,\text{food}} D_{i,p,\text{food}}(t), \quad (2.10)$$

where $\omega_{iv}(t)$ represents meal participation and serving share. The second expression shows where the single route-weight multiplication enters the total-dose accumulator; the food-pool module itself does not apply $w_{p,\text{food}}$. For norovirus, the pathogen profile assigns $w_{p,\text{food}} = 0.20$, so food contamination is not merely another linear contact edge. One infected BOH worker can seed a shared food pool that reaches many passengers who never share direct contact with that worker.

Front-of-house (FOH) staff have a different role. FOH crew move between passenger dining groups, tables, corridors, and service areas, so they act as spatial vectors between otherwise separated dining cohorts. The relevant configuration fields are the pathogen-profile `food_contamination` block and the `transmission_route_weights` entries that include the food route. The present model captures the route-level amplification and venue-level contamination pool, but it does not yet enforce full agent-level BOH/FOH segmentation with role-specific mandatory sick-leave thresholds. Chapter 10 identifies that richer crew segmentation and compliance layer as future work.

Direct contact is calculated inside each occupied zone. The contact engine inherits the shipboard

host–pathogen ABM lineage from Korkin and Srinivasan, while adding density-dependent contact scaling for the present multi-platform campaign [3, 63, 64]. In density-dependent mode, the effective number of contacts for target agent i in a zone with $N_z(t)$ occupants is sampled around

$$\bar{C}_{i,z}(t) = c_0 \left(\frac{N_z(t)}{N_{\text{ref}}} \right)^{\alpha_c} m_i(t), \quad (2.11)$$

where c_0 is the base contact rate, N_{ref} is reference occupancy, α_c is the contact-density exponent, and $m_i(t)$ includes voyage contact scaling and crew service multipliers in dining or galley zones. A hard maximum caps the draw. In legacy mode, contacts are drawn from the original average-contact pool. In heterogeneous-zone-dose mode, a mean-one lognormal exposure factor

$$H_z \sim \exp\{\mathcal{N}(-\sigma_z^2/2, \sigma_z^2)\} \quad (2.12)$$

multiplies within-zone dose so that expected mean exposure is preserved while crowded or service venues can generate stronger tails. This mechanism provides a route for offspring overdispersion without imposing a separate fixed superspreader class [65–68].

Short-range droplet exposure converts a fraction of shedding to same-room aerosol mass. The implementation uses a droplet aerosolization fraction and an inhalation fraction per epoch. For target i in zone z ,

$$D_{i,p,\text{droplet}}(t) = \phi_{\text{drop}} \psi_{\text{inh}} \sum_{j \in \mathcal{I}_{p,z}(t)} V_{j,p}(t) q_i(t) b_z, \quad (2.13)$$

where ϕ_{drop} is the droplet aerosol fraction, ψ_{inh} is the inhaled fraction, $q_i(t)$ is the confinement factor, and b_z is the balcony or zone ventilation factor. This is a same-room well-mixed approximation: the droplet term sums shedding over all infectious occupants in the zone and does not include distance-to-source, plume geometry, or within-room spatial dilution. It is appropriate for the coarse zone graph used here, but it can overstate exposure in large rooms when susceptible agents are far from the source. The same shedding mass is also added to the zone aerosol pool for later transport.

Long-range airborne exposure uses the mass present in a target zone after native or CONTAM transport. If $M_{z,p}(t)$ is aerosol mass and V_z is zone volume, then

$$D_{i,p,\text{airborne}}(t) = \psi_{\text{inh}} M_{z,p}(t) a_{\text{HVAC}}(t) b_z, \quad (2.14)$$

where ψ_{inh} is the per-occupant fraction of room aerosol mass inhaled per epoch and a_{HVAC} is the active HVAC-airborne scalar injected by protocols. The implementation therefore treats inhalation as a fraction of total room aerosol mass rather than as an explicit product of concentration $M_{z,p}(t)/V_z$ and breathing flow rate. This is a well-mixed, dilution-free approximation: it removes the explicit V_z dilution term and folds any zone-volume dependence into a single $\psi_{\text{inh}} a_{\text{HVAC}} b_z$ product. It is adequate for the platform comparisons reported here because the campaign design varies $M_{z,p}(t)$, a_{HVAC} , and b_z but not room volume ratios; it would need to be refined with an explicit V_z and breathing-flow term for absolute inhaled-dose validation or for cross-platform respiratory

comparisons that depend on room-volume contrasts.

Fomite exposure is persistent across epochs. Surface pools receive deposited mass from shedders, decay each epoch, and are picked up by later agents with a pickup probability and transfer fraction. Quarantined agents in cabin corridors skip public fomite pickup, while cabin-mate exposure remains possible. Food contamination is a separate pool for dining-type zones. A small fraction of shedding deposits into food pools and a configured fraction of that pool is ingested by agents in enabled food zones. Environmental-source transmission supports pathogens such as *Legionella* by delivering dose from a zone-scoped or global environmental reservoir even in the absence of infected human shedders [3, 30, 43, 56, 58].

2.5 CONTAM HVAC coupling

The CONTAM branch is implemented by `engines/py_contam_bridge.py` as a native Python bridge to multizone airflow and contaminant transport. CONTAM is a NIST multizone airflow and contaminant transport program, and ship-specific applications have coupled CONTAM to Wells–Riley or mechanistic aerosol risk calculations [20, 41, 69]. It reads `air_flow_paths.json`, builds zone nodes from the platform spatial layout, and constructs airflow paths from three sources: HVAC recirculation, cross-zone links, and passive adjacency. The star-topology HVAC representation sends room return air to a virtual AHU plenum and then supplies filtered recirculated air back to rooms. Outdoor-air fraction removes a fraction of return pathogen mass before supply, and filter efficiency applies to ducted supply paths.

For each zone z the bridge solves a well-mixed contaminant ODE analytically at each epoch. Inter-zone coupling is handled by updating the source term $S_{z,p}(t)$ from the upstream zones’ state at the beginning of each epoch, then integrating each zone independently under a within-epoch decoupled well-mixed assumption. This is an operator-splitting scheme, not a coupled implicit solver:

$$\frac{dM_{z,p}}{dt} = S_{z,p}(t) - k_z M_{z,p}(t), \quad (2.15)$$

$$M_{z,p}(t + \Delta t) = M_{z,p}(t)e^{-k_z \Delta t} + \frac{S_{z,p}(t)}{k_z} (1 - e^{-k_z \Delta t}), \quad (2.16)$$

where $S_{z,p}$ is the filtered inflow source rate and

$$k_z = \sum_{z'} \frac{Q_{z \rightarrow z'}}{V_z} + \lambda_p. \quad (2.17)$$

The terms $Q_{z \rightarrow z'}$, V_z , and λ_p are volumetric flow, zone volume, and natural decay or settling rate. This branch is used when airborne route attribution or native-versus-CONTAM validation is part of the experiment. For campaign-scale runs where HVAC is not the tested mechanism, the native transport branch preserves tractable runtime. This split is important because Diamond Princess analyses disagree about cross-cabin and long-range route attribution, making HVAC a scenario and

validation layer rather than a fixed assumption [20, 21, 70, 71].

2.6 Observation engine and diagnostic cascade

CTB separates infection from observation. The observation engine samples the simulated ship through instruments rather than assuming that all infections are immediately known. The instrument set includes continuous air sampling, targeted surface swabs, wastewater sequencing grids, clinical rapid diagnostics, clinical qPCR, clinical microbiology, long-read verification, wearable physiological monitoring, and fleet-level wearable summaries. Each instrument produces raw observations such as Ct values, detection booleans, anomaly scores, clinical positives, or microbiome disruption values. Those raw observations are then converted to stoplight states. Each class is anchored to an imperfect real-world instrument family: rapid antigen tests and molecular assays for respiratory viruses, clinical and environmental norovirus assays, wastewater nucleic-acid surveillance, wearable physiological anomaly detection, and sequencing or metagenomic confirmation [72–85].

A generic observation model for modality m can be written as

$$Y_m(t) = g_m\{\mathcal{S}(t - \delta_m), \theta_m\} + \epsilon_m(t), \quad (2.18)$$

where δ_m is reporting delay, θ_m contains sensitivity, specificity, coverage, and threshold parameters, and ϵ_m is instrument noise. The stoplight function is

$$Z_m(t) = h_m(Y_m(t)) \in \{\text{GREEN}, \text{AMBER}, \text{RED}\}. \quad (2.19)$$

For Ct-based instruments, low Ct values map to stronger stoplights when detection is positive. For wastewater and microbiome disruption, anomaly scores map to stoplights. For clinical RDTs, positives map directly to RED. For syndromic surveillance, the sick-call count is compared with configured AMBER and RED thresholds. These mappings are observation models, not ground truth: sensitivity, specificity, sampling cadence, reporting delay, and uptake must be retained because symptom screening and thermal screening miss presymptomatic and asymptomatic infection [53, 54, 86, 87].

The diagnostic cascade uses those stoplights to represent escalation across information sources. A virtual `detection_escalation` instrument aggregates syndromic, wearable, environmental, and clinical modes. For a set of modes $\mathcal{M}$, the integrated level can be represented as

$$L(t) = \max_{m \in \mathcal{M}} Z_m(t), \quad (2.20)$$

with the ordering $\text{GREEN} < \text{AMBER} < \text{RED}$. Protocols can require multiple affected modes, zones, or agents before firing. This design permits onboard response to depend on concordant signals rather than on a single perfect observer.

2.7 Reactive protocol engine

The reactive protocol engine is implemented in `crusher_labs/protocol_engine.py` and configured through protocol JSON files validated by schema contracts. At the end of each epoch, it recomputes stoplights from raw results, checks SOP trigger conditions, applies escalation gates, observes activation delays, activates or deactivates SOPs, merges modifiers, and debits costs through the cost ledger. This makes response a function of evidence generated by the simulated ship. The model rule is no hardcoded epoch schedules for outbreak response; interventions fire because instrument or attack-rate conditions are met.

A protocol s is eligible when its trigger condition is satisfied:

$$\mathbb{K}_s^{\text{trigger}}(t) = \mathbf{1} \left\{ \sum_{u \in U_s} \mathbf{1}[Z_{m_s, u}(t) \geq z_s] \geq n_s \right\}, \quad (2.21)$$

where m_s is the instrument class, U_s is the set of zones, agents, or modes evaluated by the trigger, z_s is the required stoplight level, and n_s is the minimum affected count. If the SOP has an escalation gate, ship status must also exceed the required rank. If it has an activation delay ℓ_s , the modifiers apply from ℓ_s epochs after the trigger first fired, rather than requiring the trigger to remain true for ℓ_s consecutive epochs. In the implementation (`crusher_labs/protocol_engine.py`), `first_seen` records the epoch at which the trigger first crossed, and the SOP is active when $t \geq \text{first_seen} + \ell_s$ and the current escalation gate $E(t) \geq E_s^{\min}$ is satisfied:

$$A_s(t) = \mathbf{1}\{t \geq \text{first_seen}_s + \ell_s, E(t) \geq E_s^{\min}\}. \quad (2.22)$$

SOP modifiers are ordinary model parameters exposed through the protocol layer. Examples include `direct_contact_scalar`, `droplet_scalar`, `hvac_filter_efficiency_override`, `close_zones`, diagnostic cadence changes, and PPE or cleaning effects. Scalar reductions use the most restrictive value when several protocols are active; list-based closures are unioned. The cost ledger records both one-time activation costs and recurring per-epoch costs. This cost accounting is important because surveillance is evaluated not only by attack-rate reduction but also by the decision quality of when expensive escalation occurs.

VSP-style escalation is represented as a threshold rule on observed illness rather than on true infection. Let $\text{AR}_{\text{cum}}^{\text{pax}}(t)$ and $\text{AR}_{\text{cum}}^{\text{crew}}(t)$ be cumulative reportable acute-gastroenteritis attack rates in passengers and crew separately. The VSP trigger is

$$\text{escalate}(t) = \mathbf{1}\{\text{AR}_{\text{cum}}^{\text{pax}}(t) \geq \theta_{\text{VSP}} \vee \text{AR}_{\text{cum}}^{\text{crew}}(t) \geq \theta_{\text{VSP}}\}. \quad (2.23)$$

The baseline calibration uses the VSP reporting threshold as a response backstop. Because the trigger is tied to observed illness, the same biological parameterization can produce a high un-

controlled attack rate and a much lower public record after response. This implements the VSP literature’s core measurement problem: 2% and 3% AGE thresholds intensify monitoring, consultation, case finding, specimen collection, isolation, and sanitation, so the public endpoint is intervention-shadowed [5, 6, 16, 88].

2.8 Overdispersion and superspreading

Superspreading is represented as an emergent distribution of secondary cases rather than as a fixed label attached to a subset of agents. The conventional summary is the negative-binomial offspring distribution

$$Z_i \sim \text{NegBin}(R, k), \quad \text{Var}(Z_i) = R + \frac{R^2}{k}, \quad (2.24)$$

where small k indicates strong overdispersion [65, 66]. For SARS-CoV-2, early cluster analyses estimated strong overdispersion, and reviews emphasize that large events arise from the concurrence of infectiousness, behavior, environment, and opportunity [66, 67]. Cruise-ship COVID-19 data show the same mixture of frequent small clusters and rare large voyages, with Diamond Princess and Ruby Princess acting as upper-tail events in worldwide cruise summaries [18, 23, 89, 90]. Norovirus cruise reconstructions similarly support heterogeneous transmission, with delayed diagnosis and isolation affecting secondary-case distributions [43, 91]. At the same time, a fitted negative-binomial k can conflate network structure, observation bias, temporal infectiousness, intervention timing, and setting mixtures [68].

ABMs capture overdispersion naturally when their contact and environment layers are sufficiently heterogeneous. In CTB, large offspring counts can arise when an infectious agent has high shedding near peak infectiousness, works as crew in a dining or galley zone, attends a crowded venue, shares a cabin, contaminates a food or surface pool, or occupies a poorly ventilated zone that feeds downstream airflow. The heterogeneous-zone-dose mode adds residual within-zone variability while preserving the mean dose. The preferred calibration workflow is therefore mechanism first and offspring-distribution check second. If the simulated zero-transmitter fraction, upper-tail secondary-case counts, largest event size, and extinction probability already match data, adding an independent superspreader multiplier would double count heterogeneity.

This point is especially relevant for cruise ships. A deterministic or homogeneous compartmental model can reproduce a mean epidemic curve while missing fade-outs and rare large outbreaks. A stratified model can separate passengers, crew, and decks, but it still averages within strata. CTB records exposure matrices and transmission events by pathway, zone, and pathogen, allowing the model to report distributions of secondary cases and route attributions rather than only final attack rates. Those outputs are the appropriate validation targets when line lists, cabin maps, venue records, environmental samples, or sequence data become available [22, 58, 71, 91].

2.9 Multi-pathogen support

Multi-pathogen support is implemented through pathogen profiles, route weights, dose-response parameters, environmental settings, food-contamination settings, and pathogen-specific infection state. The schema layer defines contracts for profiles, protocol files, run specifications, instrument parameters, decision actions, and voyage configurations. These contracts are part of the architecture: they prevent a campaign from silently changing the meaning of a field across pathogens or platforms.

For pathogen p , the profile defines a parameter set

$$\Theta_p = \{\alpha_p, \beta_p, \kappa_p, d_{\text{adj},p}, w_{p,r}, \lambda_p, \phi_{p,r}, \chi_p\}, \quad (2.25)$$

where α_p and β_p parameterize beta-Poisson dose response, κ_p parameterizes exponential response when used, $w_{p,r}$ are route weights, λ_p is environmental decay, $\phi_{p,r}$ are route-specific deposition or delivery fractions, and χ_p contains pathogen-specific clinical and environmental options. Norovirus uses contact, fomite, food, and wastewater-relevant observation. SARS-CoV-2 and measles emphasize droplet and airborne pathways, diagnostic timing, and respiratory presentation. Legionella emphasizes environmental-source and HVAC-linked exposure rather than person-to-person spread [29, 30, 54, 55, 92].

The monograph campaigns use this architecture across 4 pathogens and 4 vessel platforms. The same epoch loop, observation engine, protocol engine, and telemetry contracts are reused, but the biological parameters and route weights change. This is what makes CTB a platform for comparing surveillance and response logic rather than a single-pathogen curve-fitting exercise. The architecture also makes explicit where uncertainty remains: route-specific dose conversion, environmental signal calibration, individual compliance, and the mapping from simulated pathogen mass to measured Ct or sequencing counts. Those uncertainties are carried into later chapters as calibration and validation problems rather than hidden inside a single effective transmission rate [62, 93–96].

Chapter 3

Vessel Classes and Operational Parameterization

3.1 Purpose of the platform layer

CTB uses four synthetic vessel classes to separate ship-operating effects from pathogen biology. The platforms are not replicas of named ships. They are controlled abstractions spanning expedition, classic, spirit, and mega-cruise operating regimes. This chapter defines the modeling assumptions at a level sufficient to interpret the campaign results. Full parameter tables for zone counts, capacities, schedules, meal weights, and response blocks are reported in the appendices. The main text uses those tables only to explain why a campaign result is class-dependent. This placement reflects the review request to keep platform description secondary to inference.

The epidemiologic reason for retaining multiple platforms is that cruise-ship outbreak data are not generated by a single built environment. Passenger capacity, venue layout, cabin density, food-service format, and medical-center capacity vary with vessel class. The literature supports explicit representation of cabins, dining venues, food-service exposures, crew work networks, and regulatory observation rather than a single homogeneous mixing pool [5, 10, 50]. At the same time, the evidence does not support a simple causal rule that larger ships are always more dangerous after occupancy, voyage length, reporting, and response are considered. The platforms therefore encode plausible operating contrasts and allow calibration to test whether observed attack rates are compressed by response rather than by absence of ship-size effects [5, 88].

3.2 Four vessel classes

The expedition platform contains 450 agents. It represents small polar or specialty vessels with compact public spaces, fewer redundant venues, tighter cabin-to-public-space coupling, and lower absolute medical staffing. Dining is dominated by served meals and scheduled seating. In the

platform configuration used for the monograph campaigns (Appendix A.1, Table A.3), expedition dining is assigned zero buffet weight; buffet exposure is therefore a scenario-layer sensitivity rather than a routine mixing pathway on this class. Venue modularity is low: the same lounge, lecture, tender, and dining spaces are reused by a large fraction of the manifest. This class is therefore sensitive to repeated co-location and cabin-mate exposure even though the population is small.

The classic platform contains 1,910 agents. It represents older mid-size cruise ships with a mixture of served dining, buffet service, show lounges, bars, pools, and deck-specific cabin corridors. Compared with expedition vessels, it has more venue redundancy and a larger medical-center footprint. Compared with larger contemporary vessels, it retains more shared services and fewer highly segmented neighborhoods. The model treats the classic class as an intermediate topology in which dining schedules and cabin corridors still generate repeated contacts.

The spirit platform contains 3,000 agents. It represents a contemporary large cruise product with multiple restaurants, buffet service, specialty venues, theaters, crew-only work areas, and more separable passenger neighborhoods. Venue modularity is higher than in the classic class, so not every passenger has access to the same realized contact set each day. Buffet and casual dining occupy a larger share of meal exposures. The class also has greater response capacity in absolute terms, but that capacity must serve a larger denominator.

The mega platform contains 7,000 agents. It represents the largest resort-style vessels, where extensive venue modularity creates partial segmentation despite high onboard density. Dining is distributed across buffet, served restaurants, specialty venues, and crew food-service back-of-house flows. Cabin corridors, elevator lobbies, theater exits, and high-throughput buffets remain potential contact and fomite amplifiers, but repeated mixing is diluted by the number of alternative venues. Medical and environmental-health teams are larger in absolute terms than on smaller vessels, yet response logistics are more complex because case finding, isolation support, specimen handling, and communication must scale across a large manifest.

3.3 Dining, cabins, and venue topology

Dining topology is represented because shipboard enteric outbreaks repeatedly implicate person-to-person spread, food handling, shared dining spaces, and contaminated surfaces [12, 31, 58]. The model distinguishes served dining, buffet dining, galleys, crew messes, and food-associated environmental reservoirs. Served dining concentrates contacts within scheduled seatings and table-service staff. Buffet dining increases transient contacts, shared utensil or surface opportunities, and food-pool exposure. The platform layer assigns meal weights by vessel class rather than assuming that every ship has the same breakfast, lunch, and dinner contact structure.

Cabins are modeled as household-like compartments. This follows outbreak investigations in which ill cabin mates and cabin restriction are recurrent determinants of observed spread [10, 13, 15, 97]. Cabin density varies by class through occupancy distributions, corridor density, and the

share of agents assigned to passenger or crew accommodations. Quarantine changes the contact network but does not remove the agent from the vessel, the cabin air context, or the cabin-mate exposure set. This distinction is necessary for both norovirus and respiratory pathogens because cabin confinement can reduce public mixing while increasing within-cabin exposure.

Venue modularity captures whether the same population repeatedly uses the same public rooms. Low-modularity platforms concentrate contacts in a small set of lounges, dining rooms, and corridors. High-modularity platforms distribute passengers across restaurants, entertainment spaces, deck neighborhoods, and outdoor areas. The model uses this property to vary repeated-contact probability without converting ship class into a scalar multiplier on transmissibility. This is conservative: the ship-size review found support for explicit layers but not for a size-only causal estimate independent of operations and surveillance.

3.4 Crew, passenger, and back-of-house segmentation

Crew and passenger agents occupy different but coupled operating systems. Passenger-facing front-of-house zones include dining rooms, theaters, lounges, shops, pools, elevators, corridors, and medical-center waiting areas. Back-of-house zones include galleys, crew messes, laundry, stores, engineering spaces, crew cabins, and service corridors. The segmentation matters because crew can bridge passenger neighborhoods while also experiencing different reporting incentives, cabin density, and work schedules. Diamond Princess analyses showed that crew and food-service workers remained operationally mobile after passenger restrictions, making crew networks an explicit transmission and response layer rather than a passive population [19, 24].

The model therefore assigns role-specific mobility, dining, accommodation, and medical-response parameters. Passenger agents follow voyage schedules with meals, activities, cabins, excursions, and optional sick-call behavior. Crew agents follow work-shift and back-of-house patterns, with passenger-facing service contacts and crew-only living contacts. This design is also consistent with VSP reporting, which records passenger and crew gastrointestinal illness separately and uses thresholds that can be triggered in either population [5, 6, 9].

3.5 Medical staffing and response capacity

Medical capacity is represented as a class-specific operating constraint, not as a guarantee of perfect ascertainment. Smaller platforms have fewer clinical staff, fewer isolation or holding spaces, and less surge capacity. Larger platforms have larger medical centers and environmental-health teams, but they also face larger denominators, longer internal communication chains, and more distributed case-finding tasks. For each vessel class, the configuration specifies sick-call probabilities, detection delays, isolation support, specimen-handling capacity, sanitation effort, and response escalation blocks.

The public-health basis for this layer is the VSP system of routine reports, special reports, outbreak thresholds, sanitation inspections, and environmental-health consultation [6, 8, 9]. The model treats those rules as both an observation process and an intervention process. A case must be ill, report or be detected, meet the relevant syndrome definition, and enter the shipboard reporting stream before it contributes to threshold-driven response. After response activation, medical isolation, communication, food-service modification, enhanced cleaning, specimen collection, and public-health consultation change the exposure process that later produces the final attack rate [5, 16].

3.6 Voyage schedule, port calls, and ship-operations specification

Each campaign voyage lasts 7–14 days. The itinerary layer can place port calls inside the voyage, allowing excursions, partial passenger off-ship periods, staff workload changes, and possible introduction events to be represented. The present monograph uses port calls mainly as an operating schedule and importation hook rather than as a fully calibrated shore-side transmission model. In the campaigns reported here, port-call introductions are configuration-specified events or disabled stubs, not prevalence-driven random draws from a port-specific force of infection; no port-level prevalence series is fitted. This choice keeps the chapter focused on onboard dynamics while preserving a route for future itinerary-specific analyses.

The ship-operations specification adds three configuration blocks. First, meal weights define the class-specific distribution of agents across served dining, buffet dining, crew messes, and specialty or casual venues. Second, port-visit stubs define the timing and affected agent groups for shore excursions, passenger-turnover introductions, or altered onboard occupancy. Third, medical-response blocks define class-specific sick-call behavior, reporting delay, isolation support, diagnostic sampling, and escalation capacity. These blocks are read by the same epoch loop as the transmission, observation, and protocol engines, so operations modify exposure opportunities before and after response rather than appearing only as post hoc labels.

The modeling implication is intentionally modest. The four platforms provide a structured sensitivity frame for the monograph’s calibration and intervention chapters. They encode vessel class, dining topology, cabin density, crew–passenger segmentation, voyage duration, port-call structure, and response capacity while leaving detailed parameter tables to the appendices. This prevents platform description from dominating the main text while keeping the assumptions traceable when campaign results differ by vessel class.

Chapter 4

Pathogen Parameterization

4.1 Scientific claim

The pathogen layer is intentionally asymmetric: four pathogens define the boundary-campaign surface, while six additional profiles provide stress tests for route specificity, environmental reservoirs, and rare high-consequence introductions. Norovirus GII.4, SARS-CoV-2, influenza A, and measles were selected because cruise-ship data provide calibration anchors for enteric, respiratory, and vaccine-preventable introduction regimes. The remaining profiles, *Legionella pneumophila*, *Vibrio* spp., *Campylobacter jejuni*, *Clostridioides difficile*, Andes hantavirus, and Ebola virus, are retained as structured priors rather than as fully calibrated cruise-ship endpoints. The available calibration-target file supplies quantitative targets for SARS-CoV-2 and norovirus only; for the other pathogens, this chapter therefore records the empirical anchor and the missing calibration datum rather than inventing an attack-rate target.

4.2 Common parameter structure

Each pathogen profile maps biological evidence into the same model fields: route weights, dose response, shedding, recovery, non-susceptibility, and calibration target. The route weights classify exposures into oral ingestion, short-range respiratory or aerosol exposure, fomites, water or environmental reservoirs, and direct contact. They are not interpreted as measured fractions of every outbreak. They are priors on the relative contribution of routes before calibration against ship observations.

For pathogens with low-dose heterogeneity, the model uses a beta-Poisson family. The implemented campaign code uses the approximate beta-Poisson function stated in Chapter 2. The Teunis confluent-hypergeometric formulation below is the biological prior and sensitivity target for

norovirus, not the function that generated the $d_{\text{adj}}=10.6$ campaign outputs:

$$P_{\text{inf}}(d) = 1 - {}_1F_1(\alpha, \alpha + \beta, -d), \quad (4.1)$$

with susceptible-host estimates near $\alpha = 0.04$ and $\beta = 0.055$ for the Norwalk challenge data [59]. This is a shape prior on effective dose, not a conversion from genome copies or colony counts to infectious particles. Messner’s fractional-Poisson model and Schmidt’s review of low-dose uncertainty remain mandatory sensitivity structures for norovirus and other ingestion pathogens because aggregation, assay units, and host selection can materially alter the low-dose tail [60, 61].

For pathogens assigned an exponential dose response, infection is represented as

$$P_{\text{inf}}(d) = 1 - \exp(-rd), \quad (4.2)$$

where r is an effective per-dose infectivity parameter. This form is used for influenza A, measles, Legionella, *C. difficile*, Andes hantavirus, and Ebola. It is a tractable prior when the literature supports monotone single-hit risk but cruise-specific dose units are unavailable.

Shedding is represented as an age-since-infection process rather than a fixed infectious interval. For acute infections with a rise and decay phase, the default kernel is

$$G(a) = G_0 \exp(-\delta a) [1 - \exp(-\gamma a)], \quad \gamma > \delta, \quad (4.3)$$

with route-specific deposition from $G(a)$. Pathogens differ in whether the relevant emitted material is fecal viral RNA, respiratory viral load, water-system contamination, spores, blood or body fluid, or foodborne bacterial dose. Recovery is therefore split into clinical recovery, cessation of high infectiousness, and persistent marker positivity.

4.3 Boundary-campaign profiles

Norovirus GII.4 is the enteric calibration anchor. It is transmitted by fecal-oral contact, vomiting-associated environmental deposition, contaminated hands and surfaces, food, and water. Illness is usually short, with incubation near 24–48 hours and symptoms lasting about 2–3 days, but stool RNA can remain detectable for weeks [51–53, 98]. A longitudinal GII.4 analysis reported peak fecal concentrations spanning roughly 10^5 – 10^9 genome copies per gram and detectable shedding durations from about 8 to 60 days, while challenge reanalysis showed dose-dependent shifts in shedding and symptom timing [51, 52]. The model therefore assigns high oral and fomite weights, a lower aerosol weight restricted to vomiting plumes, and an environmental persistence component. The aerosol term for norovirus is not interpreted as an inhalation dose-response distinct from oral infection; it represents aerosolized vomitus or droplets that deposit onto hands, surfaces, or mucosa and are converted into the same effective oral-dose ledger before the beta-Poisson infection step. The non-susceptibility prior is FUT2/secretor status: about 20% nonsecretors is a practical cruise-

population default, with strong but incomplete protection against many GI.1 and GII.4 viruses [99, 100]. Calibration uses the VSP-shadowed cruise endpoint, not an uncontrolled final size: typical cruise norovirus attack rates are about 2–10% among passengers, and the calibrated CTB setting at $d_{\text{adj}}=10.6$ reproduces a narrow post-response ship-size band of roughly 5–7% while retaining a larger uncontrolled platform gradient [5, 5, 10, 16]. That gradient is a model output under the calibrated parameterization; no empirical uncontrolled cruise attack rates exist to constrain it directly, and Section 5.9 shows that alternative parameterizations along the dose–alpha ridge could reproduce the same post-response band with a different uncontrolled gradient.

SARS-CoV-2 is the respiratory beta-Poisson boundary case as implemented here, using an effective-dose approximation rather than a claim that SARS-CoV-2 biology follows the same low-dose heterogeneity model as norovirus. Transmission occurs by short-range respiratory exposure, shared-air aerosol in poorly ventilated zones, direct close contact, and contaminated surfaces, with the fomite route kept as a minor residual route rather than the main driver. Diamond Princess and other 2020 cruise outbreaks provide the calibration context. The calibration-target file identifies Diamond Princess, 712 positives among 3,711 people, as the primary large-ship response-shadowed target, an attack rate of 19.2%. This target is not unmitigated: passenger cabin quarantine and evolving response occurred during the outbreak. It is therefore used as a partially mitigated large-ship calibration context, while Greg Mortimer at 59% and Horizon at 44% are retained as high-end expedition or outlier checks [17–19, 23]. Route attribution remains uncertain. Azimi et al. estimated a large aerosol contribution, while cabin-distribution analyses argued against substantial cross-cabin HVAC spread during quarantine [20–22, 71]. The model therefore uses respiratory-dominant weights with aerosol sensitivity scenarios rather than a single fixed route fraction. Viral-load and transmission priors allow infectiousness before symptoms, substantial asymptomatic infection, and peak viral burden near symptom onset [54, 87, 101]. Non-susceptibility is not encoded as innate immunity; it is a time-varying prior from vaccination, prior infection, variant, and age.

Influenza A is the fast respiratory exponential profile. Cruise-specific outbreaks show that influenza can amplify rapidly onboard; the Sydney–Noumea outbreak produced probable influenza in 37% of responding passengers, respiratory illness in 9% of crew, 40 hospitalizations, and two deaths [27, 28]. The route weights are respiratory droplet and aerosol dominant, with a small fomite component for hands and shared touchpoints. Shedding is shorter than for norovirus: reviewed human data support an infectious window centered on a few days, with some shedding before symptoms and lower but nonzero asymptomatic shedding [55]. Recovery is represented as several days of symptomatic illness followed by a rapid decline in infectiousness. Non-susceptibility is vaccination- and strain-history dependent, not fixed host resistance. In the boundary campaign, influenza was supercritical under the tested response setting: at the largest tested introduction size, all platforms triggered and mean final attack rate was about 0.65, so calibration should use both the low- k trigger surface and cruise-outbreak final-size checks rather than only the saturated endpoint.

Measles is the rare-importation, high-transmissibility exponential profile. It is modeled as respiratory airborne dominant, with negligible fomite or waterborne contribution for ship-scale calibration.

The relevant prevalence at embarkation is usually near zero, but an exposed traveler can board during incubation and become infectious before rash [102, 103]. Ship literature shows crew involvement and susceptibility clustering: one review found 40 ship-associated measles cases, 69% among crew, and two outbreaks producing 262 secondary land cases; a 2014 cruise event affected 23 of 968 crew versus about 0.2% of passengers, with most cases having negative or unknown vaccination histories [4, 29, 104, 105]. Shedding begins before rash and ends several days after rash onset. Non-susceptibility is the dominant state variable in this profile, but vaccine efficacy at the individual level (approximately 97% for two MMR doses) is distinct from the population-level susceptible fraction. The campaign’s non-susceptibility parameter therefore combines coverage, documentation quality, prior infection, and susceptibility clustering; it is not a direct restatement of MMR efficacy. The boundary campaign encoded high non-susceptibility and used introduction size rather than endemic prevalence as the primary control variable; by the largest tested k , all platforms usually triggered but response kept mean final attack rates near 6–8%.

4.4 Additional pathogen profiles

Legionella pneumophila is the environmental respiratory profile. The exposure route is inhalation or aspiration of contaminated aerosols from potable-water, shower, spa, or cooling systems, not efficient person-to-person spread. Passenger-ship and hotel reviews, together with cruise-linked investigations, justify a persistent engineered-water reservoir with episodic aerosolization [30, 56, 106–108]. The exponential dose response is therefore applied to environmental aerosol dose. Shedding by infected hosts is not a source term. Recovery is clinical and may be prolonged, but transmission stops when exposure to the contaminated water system stops. Non-susceptibility is represented as host risk modification by age, smoking, chronic lung disease, and immune status rather than sterile immunity. The calibration target is detection of system contamination and case clusters, not secondary attack rate.

Vibrio spp. are enteric food and seawater profiles with beta-Poisson dose-response structure. The implemented route weights cover ingestion of contaminated seafood or water, fomites, and environmental-source exposure; they do not include a separate wound-exposure pathway. For *V. vulnificus*, wound exposure during marine contact is therefore an acknowledged limitation and should be represented in a future excursion or injury module rather than inferred from the current food/water route. There is no onboard person-to-person amplification term for ordinary calibration. Shedding is not treated as a major source on a cruise timescale; recovery depends on syndrome, from self-limited gastroenteritis to severe wound infection or septicemia. Host susceptibility is modified by liver disease, iron overload, immune compromise, and gastric acidity. The cruise calibration target is a foodborne or excursion-associated cluster with a shared exposure denominator rather than a propagating outbreak; the current bibliography contains only a thin *Vibrio* evidence base [31, 109].

Campylobacter jejuni is the bacterial foodborne beta-Poisson profile. Route weights place almost all exposure on contaminated food, water, or cross-contaminated kitchen surfaces, with negligible respiratory contribution. Shedding can continue after symptoms, but person-to-person transmission is inefficient compared with shared-source exposure. Recovery is usually within about a week for uncomplicated gastroenteritis, with a longer tail for severe disease and sequelae. Non-susceptibility is partial and exposure-history dependent; acid suppression and host vulnerability increase risk. Ship calibration should use attack rates among exposed diners or excursions, not all-passenger final attack rate, and should account for diagnostic under-ascertainment [31, 110–112].

Clostridioides difficile is a fomite-dominant spore profile. The model uses an exponential dose response with a large environmental persistence term and strong host susceptibility modifiers. Route weights emphasize contaminated bathrooms, cabin surfaces, medical spaces, hands, and sporidical-cleaning failure. Shedding is fecal spore shedding, with persistence after symptom improvement. Recovery is clinical response to treatment plus reduced shedding; relapse can be represented as a separate transition if the voyage length warrants it. Non-susceptibility is not a fixed immune fraction. Antibiotic exposure, age, hospitalization, proton-pump inhibitors, and prior colonization define the risk strata. Cruise-specific outbreak evidence is sparse, so the profile is calibrated as a sensitivity case using hospital and ABM priors rather than ship final-size data [37–39, 113].

Andes hantavirus is the rare contact and aerosolized-excreta profile. In ordinary hantaviruses, infection follows rodent-excreta aerosol exposure; Andes virus is unusual because person-to-person transmission has been documented. The route weights therefore combine environmental exposure and close contact, with respiratory exposure during intimate or prolonged contact as a sensitivity path. Shedding is poorly represented by routine respiratory curves, so infectiousness is modeled as a stage process tied to prodrome and cardiopulmonary disease. Recovery is prolonged and severe cases have high medical-resource demand. Non-susceptibility is unknown and is not used as a large immune fraction. The cruise anchor is the MV *Hondius* event and Andes transmission literature, which support rare-introduction stress tests rather than routine cruise prevalence calibration [33, 114–117].

Ebola virus is the direct-contact hemorrhagic-fever profile. Route weights are intended to represent blood and body-fluid contact, contaminated medical or cleaning materials, and close caregiving contact. The current profile still assigns small nonzero droplet and HVAC weights in the shared route schema; those weights should be interpreted as stress-test placeholders rather than evidence for routine airborne transmission, and the operational route for this monograph is symptom-stage body-fluid contact and infection-control failure. The exponential dose-response profile is retained for accidental exposure scenarios, including the published aerosolized-dose modeling literature, but it is not a calibrated model of routine shipboard Ebola spread [118–121]. Shedding and infectiousness rise after symptom onset; recovery includes removal from the shipboard contact network and possible prolonged viral persistence in selected body sites. Non-susceptibility is not a population-level prior. The calibration target is containment of a single imported symptomatic or incubating case, not a cruise attack-rate benchmark; environmental stability studies inform cleaning and

exposure-duration sensitivity analyses [122–124].

4.5 Parameter table

Table 4.1: Pathogen parameter profiles used for cruise-ship boundary and stress-test simulations. Route weights are qualitative priors before calibration; “target” distinguishes numeric cruise targets from qualitative anchors.

Pathogen	Class	Dose response	Dominant routes	Non-susceptibility	Calibration target or anchor
Norovirus GII.4	Enteric	Beta-Poisson; Teunis shape; fractional-Poisson sens.	Oral ingestion, fomites, vomiting deposition, food, water	FUT2 nonsecretor prior near 20%, genotype-specific residual risk	VSP-shadowed cruise AGE, typical passenger AR 2–10%; calibrated post-response band about 5–7% at $d_{\text{adj}}=10.6$
SARS-CoV-2	Respiratory	Beta-Poisson effective-dose family	Short-range respiratory, aerosol, close contact, minor fomites	Vaccination, prior infection, variant, age	Diamond Princess 19.2% partially mitigated large-ship target; Greg Mortimer 59% and Horizon 44% high-end checks
Influenza A	Respiratory	Exponential	Respiratory droplet/aerosol, minor fomites	Vaccination and strain-history dependent	Cruise outbreak checks; boundary surface saturates near 65% at largest tested k under tested response
Measles	Respiratory	Exponential	Airborne respiratory exposure	Two-dose MMR protection about 97%; susceptible pockets	Rare importation with crew susceptibility; boundary response final AR near 6–8% at largest tested k
<i>Legionella</i>	Environmental	Exponential	Aerosolized water systems, showers, spas	Host risk modifiers, no sterilizing population fraction	Water-system contamination and case-cluster detection, not secondary AR
<i>Vibrio</i> spp.	Enteric	Beta-Poisson	Seafood, water, environmental-source exposure; wound route not implemented	Liver disease, iron overload, immune compromise modify risk	Shared-source food or excursion cluster; cruise-specific numeric target absent
<i>Campylobacter jejuni</i>	Enteric	Beta-Poisson	Food, water, kitchen cross-contamination	Partial exposure-history and host-risk modifiers	Exposed-diner or excursion attack rate; all-ship target absent
<i>C. difficile</i>	Fomite-dominant	Exponential	Spores on bathrooms, cabins, medical surfaces, hands	Antibiotics, age, healthcare exposure, prior colonization	Sensitivity profile from hospital and ABM priors; cruise outbreak target absent
Andes hantavirus	Environmental, contact	Exponential	Rodent-excreta aerosol, close contact for Andes virus	Unknown; no large immune fraction	Rare-introduction stress test anchored to Hondius and Andes transmission literature
Ebola virus	Contact	Exponential	Body fluids, caregiving, contaminated materials; small schema HVAC weight treated as placeholder	None encoded as population immunity	Single-case containment and infection-control stress test, not cruise AR

4.6 Calibration implications

The boundary campaign uses introduction size, route weights, and non-susceptibility as separate axes because the same final attack rate can be produced by different mechanisms. Norovirus demonstrates threshold compression by VSP response. SARS-CoV-2 demonstrates route-attribution ambiguity in a well-observed respiratory outbreak. Influenza demonstrates rapid saturation when the response surface is already supercritical. Measles demonstrates that high intrinsic transmissibility can coexist with low final attack rate when non-susceptibility is high and introductions are rare.

The six additional profiles are included as structured stress tests rather than as statistical guards against overfitting. They force explicit handling of engineered-water reservoirs, shared-source food-borne exposure, spore persistence, rare zoonotic introductions, and body-fluid contact. Because the supplied calibration-target file does not provide numeric attack-rate targets for these six pathogens, their parameters should be reported as priors with sensitivity ranges until ship-specific outbreak denominators, exposure histories, or environmental measurements are available.

Chapter 5

Calibration and the VSP Shadow

5.1 Scientific claim

Cruise-ship outbreak records are post-intervention measurements, not direct samples from an uncontrolled epidemic process. For norovirus, the Vessel Sanitation Program (VSP) threshold defines both when an outbreak enters the observed data stream and when the operating response changes the epidemic trajectory. Calibration must therefore target a coupled system of infection, reporting, and response. At $d_{\text{adj}}=10.6$, CTB reproduces a narrow observed norovirus attack-rate band while retaining a larger uncontrolled platform gradient; this is possible because the VSP response compresses final sizes after detection rather than because vessel architecture is epidemiologically irrelevant [5, 6, 10, 16].

5.2 Why intervention-shadowed calibration is hard

Calibration is usually posed as an inverse problem: choose a parameter vector θ such that model summaries $m(\theta)$ match empirical summaries y . That framing is incomplete when the empirical data are generated after an adaptive response. For VSP acute gastroenteritis records, illness accumulation, sick-call behavior, sanitation, case restriction, food-service changes, and reporting thresholds are part of the observation system [5, 5, 6, 8–10, 50, 57, 88, 125]. The correct comparison is therefore not

$$y_{\text{VSP}} \approx A_{\text{uncontrolled}}(\theta), \quad (5.1)$$

but

$$y_{\text{VSP}} \approx \text{Report}(A_{\text{post}}(\theta, \rho, \tau)), \quad (5.2)$$

where A_{post} is the post-response final cumulative infection attack rate, ρ denotes the reporting and care-seeking process, and τ denotes the thresholded response rule. The $\text{Report}(\cdot)$ mapping converts infections into reported syndromic AGE by combining the pathogen-specific symptomatic

fraction, care-seeking probability, VSP-eligible symptom criteria, and reporting window. In this chapter, “attack rate” therefore refers to the reported syndromic AGE rate under Report when it is compared with VSP records, and to the underlying cumulative infection rate when it is used as a simulated outcome without observation. The difference is not semantic. A model forced to match VSP records as uncontrolled final sizes will understate biological or contact-mediated transmission on vessels where the response would have truncated spread. This is the same measurement problem seen in published outbreak investigations: the reported final count is produced after medical-center reporting, isolation, enhanced cleaning, specimen collection, passenger communication, and public-health consultation have already changed exposure opportunities [5, 13, 16, 126].

The CTB implementation used here is the open-source model in the [Crusher to the Bridge repository](#). The chapter treats the simulations as a calibrated observation model for outbreak data rather than as a claim that any single vessel class has a fixed intrinsic attack rate.

The repository and simulations are therefore used here as an observation-process model. The empirical object being fitted is not simply a pathogen’s intrinsic growth rate; it is the distribution induced by cruise operations, syndromic ascertainment, regulatory thresholds, and response.

5.3 VSP regulatory history and the 3% threshold

The VSP began in 1975 after repeated gastrointestinal outbreaks on cruise ships in the early 1970s. Its authority rests on the Public Health Service Act and foreign-quarantine regulations, including 42 U.S.C. §264 and 42 CFR Part 71, which authorize federal measures to prevent communicable-disease introduction and require gastrointestinal-illness reporting by covered vessels [5, 127, 128]. The program has kept shipboard diarrheal-illness records since April 1, 1975 and applies primarily to passenger vessels entering the United States from foreign ports with at least 13 passengers [5, 6].

The Operations Manual converts that authority into an auditable operating system. Archival editions include 1989, 2000, 2005, 2011, and 2018 manuals, with separate construction guidance for vessel design and renovation [6, 129]. Case definitions changed over time: the original 1975 reporting definition focused on at least three watery stools in 24 hours; the 2000 manual broadened the definition to include loose stools or vomiting plus another qualifying symptom; and the 2011 revision aligned diarrhea with frequency above normal for the individual [5, 6]. Those changes improved surveillance sensitivity but created nonstationarity in the longitudinal series, which is why calibration cannot treat all VSP-era counts as identically measured outcomes [5, 8, 50].

The current reporting ladder has three operational levels. A routine cumulative AGE report is due 24–36 hours before arrival at a U.S. port, with a four-hour update if counts change. At 2% cumulative AGE in passengers or crew, the vessel submits a special report and VSP begins active monitoring. At 3% or more in either population, the event is classified as an outbreak and VSP ordinarily provides intensified epidemiologic and environmental consultation or conducts an onboard investigation when feasible [5, 6, 16]. The reviewed historical sources document the 3%

rule as established VSP practice but do not identify a single archival promulgation date or a formal validation study selecting exactly 3%; the defensible interpretation is an operational threshold for surveillance, resource allocation, and public reporting rather than a biological discontinuity in norovirus transmission [5, 88, 130].

The VSP data products are therefore selective. Public outbreak reports commonly include ship, line, voyage dates, passenger and crew denominators, reported AGE counts, predominant symptoms, etiologic agent when confirmed, and response measures, but they omit unreported illness, non-threshold voyages, events outside jurisdiction, and cases outside the 15-day surveillance window for long voyages [5, 6]. MIDRS received 37,258 voyage reports from 252 ships during 2006–2019, but CDC characterized incidence estimates as underestimates because only medical-center presentations meeting the syndromic definition are counted [5, 5]. The public VSP record is thus a threshold-selected, clinic-ascertained, jurisdictionally censored dataset.

Internationally, VSP is one component of a broader maritime health regime. The International Health Regulations require surveillance and response capacities at ports, Maritime Declarations of Health, and ship sanitation certificates, while WHO’s ship-sanitation guidance provides technical control measures [7, 131]. EU SHIPSAN harmonizes passenger-ship hygiene, inspection, and communicable-disease practices across participating European authorities, and industry standards such as CLIA protocols can activate company SOPs earlier than public thresholds [4, 132, 133]. These systems are not interchangeable with VSP: thresholds, jurisdictions, reporting windows, public posting practices, and intervention rules differ, so pooled calibration data require an explicit observation model.

5.4 Norovirus dose response and shedding model

Norovirus is a useful calibration anchor because its biological uncertainties are explicit. Infection can occur after a small oral dose; clinical illness is usually short, but fecal viral RNA can remain detectable well beyond symptom resolution [51–53, 59, 98]. Symptoms, shedding, and infectiousness should therefore be modeled as related but separable states. Population and cruise reviews also show that norovirus burden is high enough for repeated introductions but strongly shaped by season, genotype, cabin exposure, food-service exposure, and reporting behavior [10, 92, 134–136]. For a host i , CTB represents an ingested effective dose $d_i(t)$ accumulated from route-specific contacts and environmental reservoirs. The Teunis dose-response model gives infection probability

$$P_{\text{inf}}(d_i) = 1 - {}_1F_1(\alpha, \alpha + \beta, -d_i), \quad (5.3)$$

with susceptible-host estimates near $\alpha = 0.04$ and $\beta = 0.055$ in the Norwalk challenge analysis [59]. Equation 5.3 is used as a biological prior on shape, not as a direct conversion from RT-qPCR genome copies to infectious particles. The fitted d_{adj} parameter absorbs that effective-dose conversion plus residual route-scale uncertainty.

The shedding kernel separates acute contamination from longer post-symptomatic detectability. A convenient rise-and-decay form is

$$C_i(a) = C_{0i} e^{-\alpha_s a} \left(1 - e^{-(\beta_s - \alpha_s) a}\right), \quad \beta_s > \alpha_s, \quad (5.4)$$

where a is time since infection and $C_i(a)$ is fecal shedding intensity. Empirical longitudinal analyses report peak concentrations spanning roughly 10^5 – 10^9 genome copies per gram and detectable shedding durations from about 8 to 60 days in GII.4 outbreaks [51]. Challenge reanalysis shows that higher inoculum can shift shedding and symptom timing earlier, while symptomatic and asymptomatic infections can have overlapping fecal viral burdens [52, 137]. Those values do not imply that all PCR-detectable material is equally infectious. They justify a model in which vomiting and acute fecal shedding generate high-amplitude deposition, while late shedding contributes a lower and more uncertain environmental source.

Route-weighted force of infection is represented as

$$\lambda_i(t) = s_i \sum_{r \in \mathcal{R}} w_{p,r} \left[\sum_{j \neq i} \kappa_{ijr}(t) g_j(t) + \eta_{ir}(t) W_r(t) \right], \quad (5.5)$$

where s_i is host susceptibility, $w_{p,r}$ is the pathogen-specific route weight for route r , κ_{ijr} is effective person-to-person contact, $g_j(t)$ is infectious shedding, $W_r(t)$ is environmental contamination, and η_{ir} is route-specific exposure to that reservoir. Equation 5.5 connects to the dose-response step through the total accumulated dose $D_{i,p}^{\text{total}}(t) = \int^t \lambda_i(u) du$ in the implementation’s per-epoch accumulator; the infection probability in Equation 2.7 or its exponential counterpart is then evaluated on $D_{i,p}^{\text{total}}(t)$ rather than directly on $\lambda_i(t)$. For norovirus, the important point is that $w_{p,r}$, s_i , contact opportunity, and dose scale can compensate in aggregate outcomes unless some of them are externally fixed. For that reason, the norovirus profile treats the Teunis dose-response equation as a shape prior, evaluates fractional-Poisson and effective-dose alternatives as sensitivity structures, and keeps host non-susceptibility separate from dose scaling [59–61, 99, 100].

5.5 The VSP threshold as observation rule and intervention trigger

The VSP threshold operates on reported acute-gastroenteritis burden. Let $G_P(t)$ be cumulative passenger cases meeting the reporting definition and N_P the passenger population. A thresholded response is triggered when

$$\frac{G_P(t)}{N_P} \geq \theta_{\text{VSP}}. \quad (5.6)$$

Equation 5.6 is simultaneously a measurement rule and an intervention rule. Before the threshold, mild infections, non-reporting passengers, and preclinical spread can remain partly hidden. After the threshold, sanitation, isolation, communication, and operational restrictions change the

transmission process that generates the final attack rate [5, 6, 13, 16, 97].

The threshold sweep shows progressive compression. Without response, platform mean attack rates span 10.6–23.4%. This gradient is a model output under the calibrated parameterization; no empirical uncontrolled cruise attack rates exist to constrain it directly. The identifiability experiment (Section 5.9) shows that alternative parameterizations along the dose–alpha ridge could reproduce the same post-response band with a different uncontrolled gradient. Sick call without the regulatory backstop reduces the range to 7.3–13.2%. VSP thresholds at 5%, 3%, and 1% reduce the corresponding ranges to 5.8–7.9%, 5.1–7.0%, and 4.3–6.0%. The same biological setting can therefore look like a high-gradient epidemic process before response and a narrow empirical distribution after response. This compression is consistent with the VSP literature’s warning that posted outbreak records are neither all-voyage incidence nor all-infection incidence; they are final syndromic counts generated after threshold-triggered changes in ascertainment and control intensity [5, 5, 10, 88].

Medical response parameters did not materially change final norovirus attack rates once the VSP backstop activated. In the tested range, sick-call probability, detection delay, and compliance were absorbed by the thresholded escalation process. This does not imply that care-seeking or compliance never matter. It means that final attack rate, under a strong VSP backstop, is a poor endpoint for identifying those upstream parameters. Time to detection, cabin restriction burden, sanitation cost, and passenger disruption may remain more sensitive endpoints. Published norovirus cruise investigations support this endpoint shift because delayed reporting, cabin-mate exposure, passenger behavior, and rapid isolation can change outbreak size even when final attack-rate records are already threshold-compressed [13, 15, 91, 97].

5.6 Multi-platform norovirus calibration

At $d_{\text{adj}}=10.6$, the no-response condition produces a physical platform gap: 23.4% attack rate on expedition, 22.6% on classic, 12.2% on spirit, and 10.6% on mega. Under the platform-default syndromic/VSP-like response batch, the same setting compresses to 6.7%, 7.1%, 4.6%, and 5.5%, respectively; the separate threshold sweep in Section 5.5 used a different factor-at-a-time robustness batch and gives a 5.1–7.0% band at the 3% threshold. The calibrated parameter is therefore not a claim that the uncontrolled epidemic is flat across ships. It is a claim that the observed VSP-scale endpoint is flat after the VSP-like response has acted.

The boundary Stan fits support this interpretation. For norovirus, Stage A contained 2,400 runs and 1,976 threshold-triggered events, an outbreak fraction of 82.3%. The reported platform summaries are inverse-logit-transformed marginal fitted probabilities, not raw logit coefficients or literal $k = 0$ intercepts: expedition 0.82, mega 0.81, spirit 0.81, and classic 0.85 at the observed design points. Stage B was fit on the trigger-positive runs and estimated lower attack-rate intercepts for the larger spirit and mega platforms than for expedition and classic. The two-stage

structure matters: outbreak occurrence and conditional final size are different summaries, and VSP-like response acts most directly on the latter. Cruise epidemiology provides the same distinction qualitatively: ship size, voyage duration, cabin sharing, crew continuity, and reporting thresholds influence whether an event is observed and how large it appears after response [4, 5, 10, 50].

5.7 Multi-pathogen boundary surfaces

The new boundary campaign generalizes the calibration argument beyond norovirus. The calibration targets reflect distinct empirical histories: norovirus VSP AGE reports, Diamond Princess and other cruise SARS-CoV-2 outbreaks, influenza cruise outbreaks, measles ship importation and crew susceptibility, and rare environmental or emerging hazards such as Legionella and Andes hantavirus [5, 17, 18, 23, 27–30, 33]. For each pathogen and platform, the outbreak surface reports $P_{\text{trigger}}(k)$, the probability that a threshold response is triggered at introduction size k , and $E[A(k)]$, the expected final attack rate. Each platform–pathogen– k cell in the boundary campaign contains 100 replicate runs for norovirus or 200 replicate runs for SARS-CoV-2, influenza, and measles. The full campaign comprises 10,000 core runs across 4 pathogens, 4 platforms, and 6–8 k values per pathogen; Appendix A.4 reports the expanded 17,600-run ledger including the additional sparse high- k cells used for influenza and measles surfaces. The surfaces are monotone except for one trivial SARS-CoV-2 fluctuation on the classic platform, where P_{trigger} and $E[A]$ at $k = 4$ are slightly below $k = 3$.

Table 5.1: Boundary-surface endpoints at the largest tested introduction size. Values are means across replicated runs.

Pathogen	Platform pattern at largest k	k_{max}	P_{trigger}	$E[A]$	P_{accel}
Norovirus	all platforms trigger with moderate final sizes	7	0.99–1.00	0.145–0.179	0.15–0.51
SARS-CoV-2	expedition triggers almost always; large ships less often	7	0.31–1.00	0.079–0.259	0.09–0.99
Influenza	all platforms trigger and reach high final size	10	1.00	0.650–0.651	0.86–1.00
Measles	all platforms usually trigger, with low mean final size under response	15	0.945–1.00	0.061–0.080	0.34–1.00

The Stage A Stan summaries separate three regimes. Norovirus is a thresholded enteric process: 1,976 triggers in 2,400 runs, with platform differences in trigger probability but substantial compression of final sizes. SARS-CoV-2 is boundary-sensitive: 1,433 triggers in 4,800 runs, with design-marginal trigger probabilities of 17.0% on classic, 19.1% on spirit, 20.3% on mega, and 63.0% on expedition. These summaries are conditional on the shared model priors and do not include a platform $\times$ k interaction term; observed raw outbreak fractions vary across cells and are reproduced in Appendix A.5. Influenza is supercritical under the tested conditions: 4,660 triggers

in 4,800 runs, and the surface reaches $P_{\text{trigger}} = 1$ with $E[A] \approx 0.65$ on every platform by $k = 10$. Measles is dominated by introduction and immunity geometry: 3,548 triggers in 5,600 runs, with the expedition platform having the highest design-marginal trigger probability at 74.7%. This interpretation matches the ship literature, where measles risk is linked to travel importation, crew vaccination documentation, and susceptible pockets rather than to continuous endemic circulation onboard [4, 29, 102, 103].

These surfaces show why a single attack-rate target is insufficient. A response threshold can suppress conditional final size for norovirus and measles while leaving influenza near saturation. For SARS-CoV-2, the same endpoint can reverse apparent platform ordering depending on whether the analysis conditions on triggering, acceleration, or final attack rate. In this monograph, $P_{\text{accel}}(k)$ denotes the estimated probability that the epidemic doubling time within a run is at least halved after the first VSP-eligible detection relative to the pre-detection interval, using the run-level trajectory stored in `outbreak_surface.csv`. It is a decision-relevant summary of trajectory shape rather than a claim about biological recovery. Calibration should therefore fit outbreak probability and conditional attack rate jointly, and should report how conclusions change across k , platform, and response threshold. The same logic applies to respiratory route attribution: Diamond Princess analyses estimate different aerosol, droplet, fomite, and cabin-transmission contributions, so a single final attack rate cannot identify route weights without environmental or contact data [20–22, 71].

5.8 Parameter identifiability

The identifiability problem must be stated formally. Let

$$\theta = (\lambda, z_0, s, n_{\text{eff}}), \quad m(\theta) = (A(\theta), P_{\text{out}}(\theta)), \quad (5.7)$$

where λ is an effective transmission scale, z_0 is the introduction-boundary condition, s is a susceptibility or non-susceptibility coordinate, n_{eff} is an effective dispersion/contact-density coordinate, A is final attack rate, and P_{out} is outbreak probability. If all four components of θ are free and calibration uses only these two scalar summaries, then the local sensitivity matrix

$$J(\theta) = \frac{\partial m}{\partial \theta} \quad (5.8)$$

is 2×4 , so $\text{rank}(J) \leq 2 < 4$. Four interacting parameters cannot be uniquely identified from two aggregate constraints. This is the structural limitation emphasized in epidemic-model calibration guidance: goodness of fit is not the same as parameter identifiability, and uncertainty must be propagated into forecasts and policy comparisons [48, 49, 62, 93, 94].

The practical defense is not that the inverse problem becomes identified by simulation volume. Repeated runs reduce Monte Carlo error around the same summaries; they do not create independent information directions. The practical reduction is therefore conditional, not a proof of

identifiability. In the reported CTB fits, susceptibility is fixed or tightly externally informed by FUT2 secretor status and seroprevalence, and the dose-response shape parameters are fixed from challenge-study evidence through Equation 5.3. Those constraints leave an effective transmission scale and an introduction-boundary or dispersion coordinate as the main fitted directions, but they do not make the full four-coordinate inverse problem uniquely identified. Even then, the result should be reported as a compatible region, not as a unique mechanistic estimate. Bayesian and likelihood-free calibration can regularize the problem with priors and simulation summaries, but prior-induced concentration is not the same as data identifiability; posterior contraction and synthetic recovery are required diagnostics [95, 96, 138].

The sensitivity interpretation follows from the same rank argument. Attack rate mainly constrains an effective transmission scale. Outbreak probability adds information about stochastic emergence and offspring heterogeneity, often represented by an offspring-dispersion parameter [65, 139]. Separate susceptibility, contact number, route weights, and dose conversion remain confounded because they enter the force of infection multiplicatively in Equation 5.5. The reported $d_{\text{adj}}=10.6$ value is therefore a calibrated effective-dose scale conditional on the chosen biology and response model, not an independently identified physical virion-to-infection conversion factor.

5.9 Parameter identifiability: synthetic recovery experiment

A 1,200-run synthetic recovery experiment tested whether the calibrated dose scale and density-dependent contact exponent could be recovered from simulated data generated near, but not entirely on, the $d_{\text{adj}} \times \alpha_c$ compensating ridge. Not all six generating vectors lie exactly on the ridge; the design deliberately includes off-ridge points to expose whether the identifiability failure is confined to the ridge or extends into its neighborhood. The design used six parameter vectors and four platforms, with 50 replicate runs per vector–platform cell. The statistical analysis followed the two-stage calibration logic used elsewhere in this chapter: a pooled outbreak-occurrence model for threshold crossing, a pooled attack-rate model for final size, and a latent recovery step for $(d_{\text{adj}}, \alpha_c)$.

Outbreak occurrence alone did not identify either coordinate on the ridge. In the pooled Bernoulli outbreak model, the dose slope was near zero, with $\beta_d = -0.033$ and a 90% credible interval of $[-0.269, 0.194]$. The contact-density slope was likewise unidentified, with $\beta_\alpha = 0.020$ and a 90% credible interval of $[-0.859, 0.911]$. In rounded terms, the outbreak model gave β_d CI $[-0.27, 0.19]$ and β_α CI $[-0.86, 0.91]$, so neither d_{adj} nor α_c was identified from outbreak occurrence alone.

Attack rates partially broke the ridge. In the pooled Beta attack-rate model, $\beta_d = 0.364$ with a 90% credible interval of $[0.259, 0.469]$, so d_{adj} was identified from final-size contrasts across generating vectors. The positive β_d is a dose-scale slope in the two-stage regression on final attack rate, not a recovery-rate parameter of the underlying epidemic; recovery dynamics enter the model separately through the shedding kernel in Equation 5.4. When the attack-rate model was fit to outbreak runs only, the dose signal strengthened to $\beta_d = 0.569$ with a 90% credible interval of

$[0.49, 0.65]$. The α_c slope remained unidentified in all models; every credible interval for β_α covered zero.

The latent recovery diagnostic showed the same structure. All six parameter vectors converged to the same posterior region, with recovered medians near $d_{\text{adj}} \approx 10.8\text{--}10.9$ and $\alpha_c \approx 0.77$, despite true values spanning $d_{\text{adj}} = 9.5\text{--}11.5$ and $\alpha_c = 0.55\text{--}1.0$. The true coordinates fell inside broad 90% intervals, but the posterior did not separate the six generating vectors. The experiment therefore supports a constrained interpretation: d_{adj} is an effective composite parameter that can be estimated from attack-rate information, but d_{adj} and α_c are not jointly identifiable from the aggregate summaries used here. The two-stage model partially breaks the ridge through final-size contrasts, not through outbreak indicators alone.

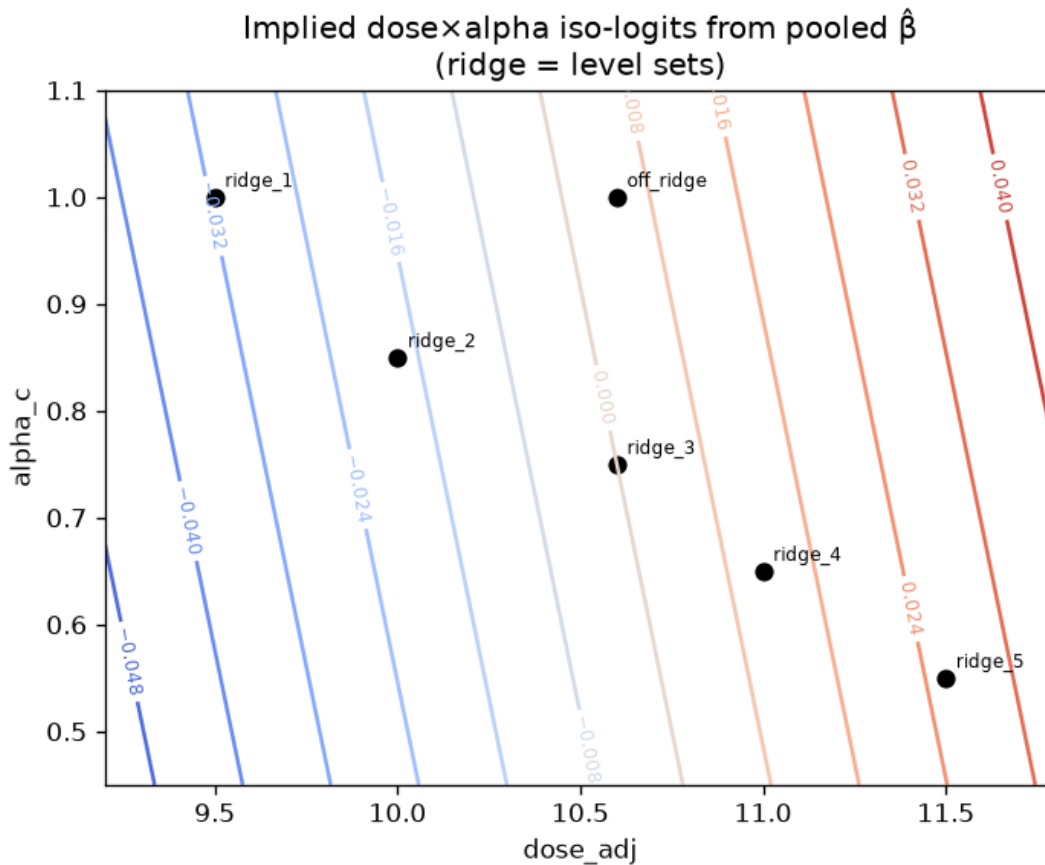


Figure 5.1: Synthetic-recovery ridge geometry across d_{adj} and α_c . Iso-logit contours show why outbreak occurrence supplies weak information about the two coordinates when the design lies along a compensating ridge.

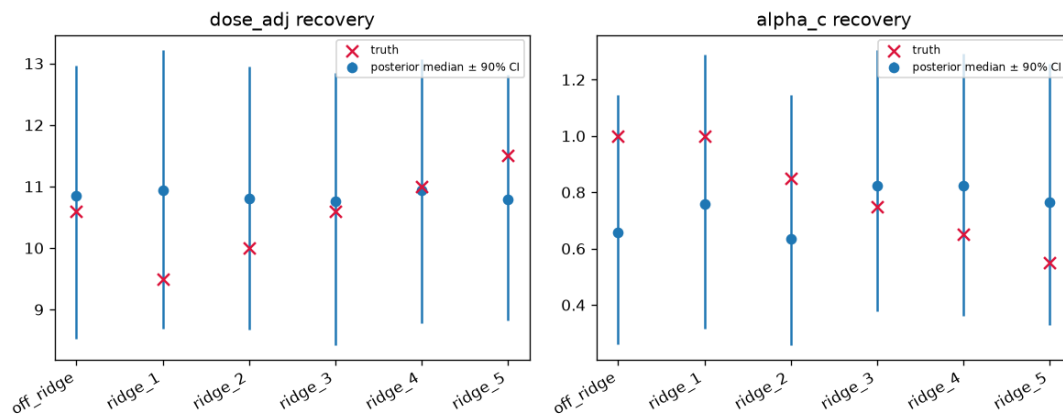


Figure 5.2: Latent recovery of d_{adj} and α_c for the six synthetic parameter vectors. Posterior medians concentrate near a common region rather than recovering distinct true coordinates, indicating joint non-identifiability under the aggregate summaries.

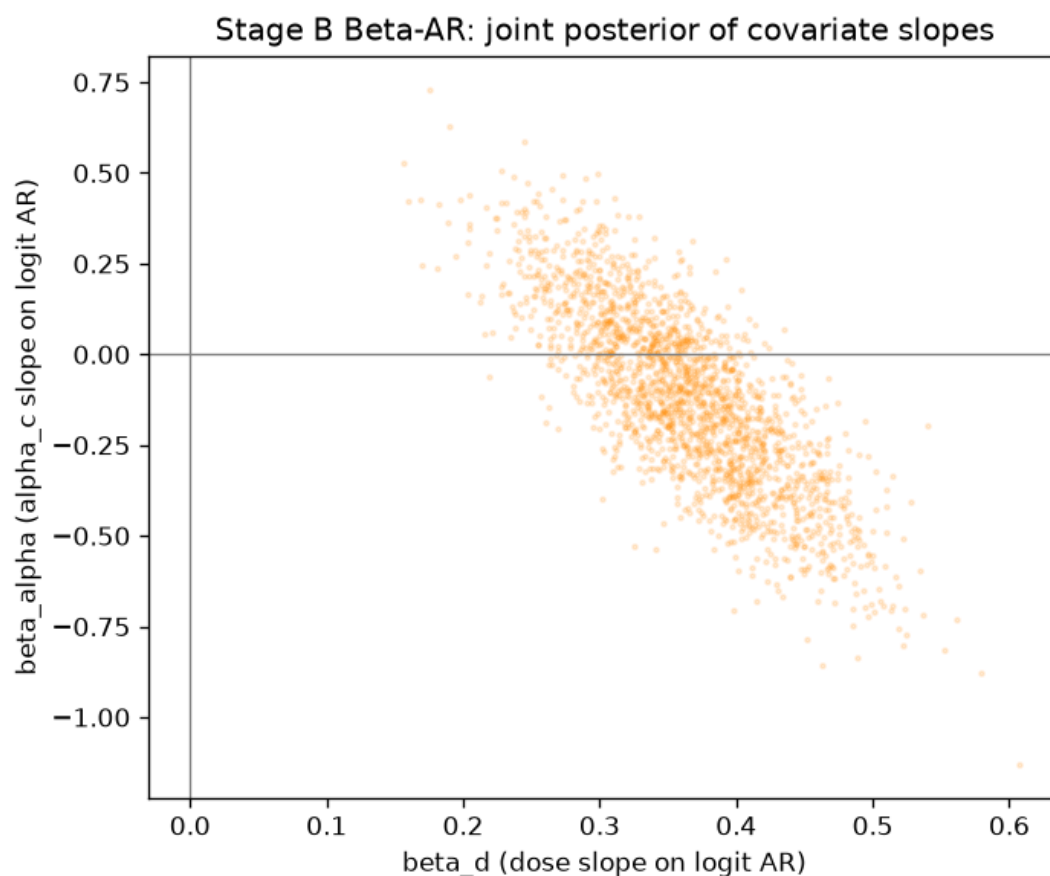


Figure 5.3: Joint posterior for the pooled attack-rate slopes. The d_{adj} slope is separated from zero, while the α_c slope remains compatible with zero, consistent with partial ridge-breaking by attack-rate contrasts.

5.10 VSP shadow stress test

A 6,360-run VSP degradation sweep tested whether the response shadow remains robust when operational parameters are weakened. The sweep varied the VSP threshold, detection delay, isolation compliance, and sick-call probability. The analysis defined the VSP shadow as broken when the absolute expedition–mega attack-rate gap exceeded five percentage points:

$$|A_{\text{expedition}} - A_{\text{mega}}| > 0.05. \quad (5.9)$$

The factor-at-a-time sweeps did not break the shadow. The maximum expedition–mega gap was 3.6 percentage points when detection delay was swept, 4.4 percentage points at isolation compliance of 0.3, 3.1 percentage points when sick-call probability was swept, and 4.2 percentage points when the VSP threshold was swept. Thus, no single-parameter degradation produced a gap above the five percentage-point criterion.

The threshold–compliance interaction was the first factor-pair setting that broke the shadow. At VSP threshold 0.20 and isolation compliance 0.1, expedition attack rate was 19.08% and mega attack rate was 26.05%, a signed gap of -7.0 percentage points. The other eight threshold–compliance cells remained within the criterion. The delay–sick-call interaction never broke the shadow; the maximum gap was 3.2 percentage points. In the worst-case panel, 9 of 27 cells broke the five-point criterion. The smallest absolute broken gap was 5.1 percentage points at VSP threshold 0.30, detection delay 12 epochs, and isolation compliance 0.3; the lowest degradation score among broken cells was 1.14 at VSP threshold 0.20, detection delay 4 epochs, and isolation compliance 0.1, with an absolute gap of 6.4 percentage points.

The direction of breakage is informative. When the VSP shadow failed, the platform gradient flipped: mega-class ships had higher attack rates than expedition-class ships. This is the opposite of the uncontrolled norovirus gradient reported earlier in the chapter. Under weak response, dense contact geometry on mega-class ships overwhelms the response system and produces higher final burden. The stress test therefore reinforces rather than weakens the calibration claim, while remaining a stress-test result rather than a standalone validation of spatial fidelity. Realistic operational variation preserves the VSP shadow, while extreme degradation exposes the underlying interaction between contact topology and response effectiveness.

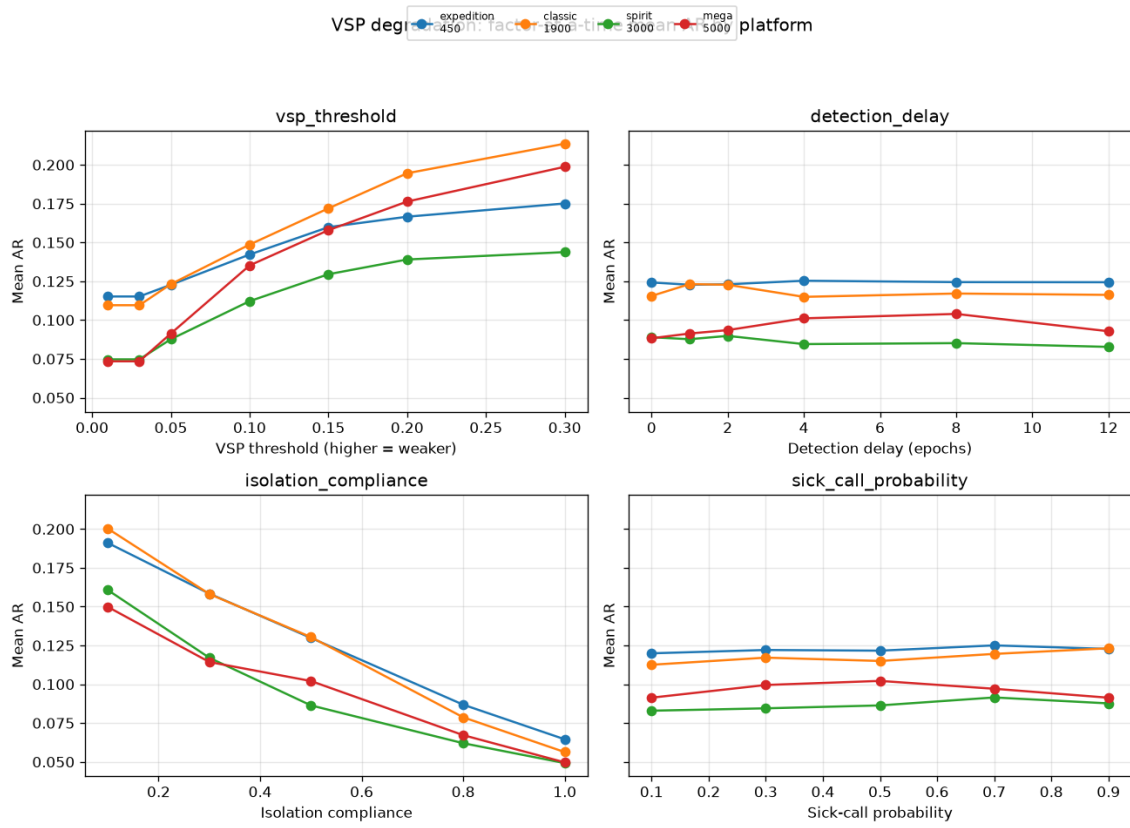


Figure 5.4: Factor-at-a-time VSP degradation sweeps. None of the single-parameter sweeps exceeded the five percentage-point expedition–mega gap criterion for breaking the VSP shadow.

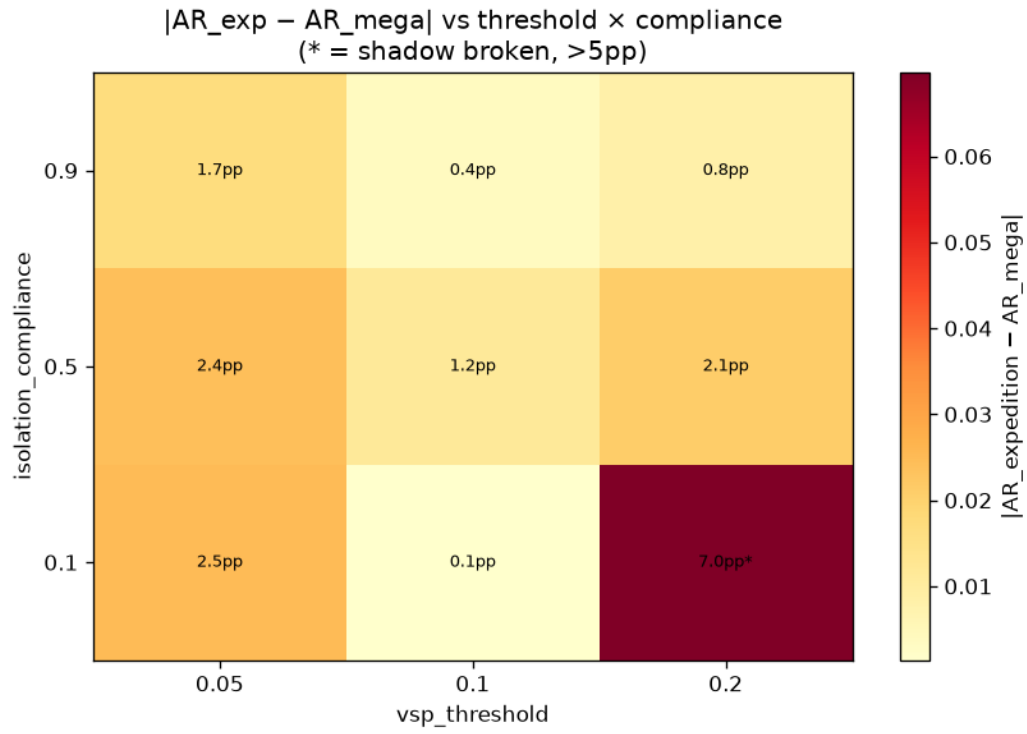


Figure 5.5: Threshold–compliance interaction. The VSP shadow broke only under the joint degradation of a 20% VSP threshold and 10% isolation compliance, where mega-class attack rate exceeded expedition-class attack rate.

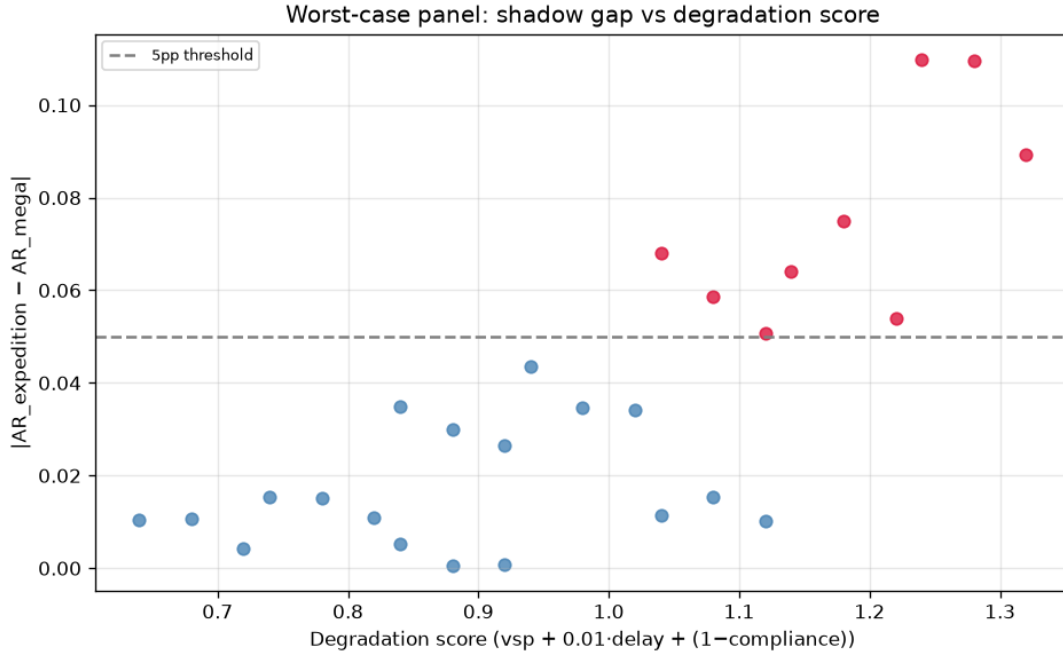


Figure 5.6: Worst-case degradation panel. Nine of 27 cells broke the five percentage-point criterion, with the smallest broken gap at 5.1 percentage points and the lowest broken-cell degradation score at 1.14.

5.11 Genotype-specific severity and susceptibility

Norovirus genotype matters for severity, host susceptibility, and possibly response-trigger timing. FUT2 controls expression of secretor-associated histo-blood-group antigens, and a practical cruise population prior is that roughly 20% of passengers are nonsecretors. Nonsecretors are strongly protected against many GI.1 and GII.4 infections, but protection is not absolute and is not uniform across genotypes [98–100, 140]. A model that treats 20% of passengers as universally immune to all norovirus genotypes would confuse host susceptibility with genotype-specific binding and pre-existing immunity.

GI.4 has been associated with more severe outcomes than pooled non-GII.4 outbreaks in several settings, but severity ratios are not transmission ratios [141–143]. A higher hospitalization or symptom-duration burden can increase reporting probability and operational response even if the infection attack rate is unchanged. The directly relevant cruise-ship comparison is Preston et al.’s GII.4-versus-GII.17 outbreak analysis for September 2020 through March 2025 [144]. Older cruise molecular epidemiology shows mixed strains and genotype replacement across seasons, so Preston’s cruise-specific GII.4/GII.17 comparison should supersede extrapolation from pediatric or healthcare settings once full tables are available [57, 145–147]. The available bibliographic summary does not provide enough numerical detail to assign a defensible GII.17 multiplier in this chapter. The correct model use is to estimate genotype-specific symptom, reporting, duration, and response-

shadowed reported-attack-rate effects from those outbreak data once the full table is available; these effects should not be interpreted as portable biological multipliers unless estimated inside a genotype-specific observation-and-response model.

This genotype layer reinforces the VSP-shadow argument. A genotype with greater vomiting, longer illness, or higher medical presentation can cross the VSP threshold earlier and receive stronger response at a lower true infection burden. The observed final attack rate may therefore decrease, remain flat, or increase depending on how biological severity changes the detection and response process. For CTB calibration, genotype-specific severity should enter both the natural-history model and the observation model. It should not be folded silently into d_{adj} .

5.12 Implications for CTB calibration

The calibration target for cruise outbreaks is a joint distribution over outbreak occurrence, threshold triggering, conditional attack rate, timing, and response burden. Norovirus at $d_{\text{adj}}=10.6$ satisfies the central calibration requirement because it reproduces the post-response VSP-scale band while preserving uncontrolled platform heterogeneity. The multi-pathogen boundary surfaces show that this reconciliation is pathogen-specific: influenza saturates the response system, SARS-CoV-2 is introduction- and platform-sensitive, measles is shaped by immunity and introduction geometry, and norovirus is strongly compressed by thresholded sanitation response.

The practical rule for the rest of the monograph is therefore simple. Do not calibrate CTB to VSP records as if they were uncontrolled final sizes. Calibrate to the mechanism that produced the records: exposure, infection, symptoms, reporting, threshold-triggered intervention, and post-response final size. That framing is more restrictive than a free fit to attack rate alone, and it makes explicit which parameters are biologically fixed, which are externally informed, and which remain weakly identifiable from aggregate outbreak data [5, 10, 62, 94].

Chapter 6

Transport Solver Validation

6.1 Why CTB keeps two transport solvers

Crusher to the Bridge (CTB) uses two implementations of airborne and environmental transport because the monograph asks two different classes of question.

The native analytical solver is the campaign engine: it is deterministic within an epoch, fast enough for large factorial sweeps, and exposes a compact set of parameters that can be varied with pathogen, vessel class, and response protocol.

The CONTAM branch is the physical-reference engine: it represents multizone airflow, pressure-driven exchange, supply and return paths, outdoor-air fractions, and contaminant transport using the National Institute of Standards and Technology CONTAM framework [69].

The validation question is therefore not whether the two codes compute identical environmental mass in every zone.

It is whether solver choice changes the epidemiological state variables used for intervention comparison, calibration, and cost-effectiveness claims.

This distinction follows the structure of prior ship and built-environment models. CONTAM has been used directly for cruise-ship airborne risk assessment, often with Wells–Riley infection calculations rather than autonomous agents [41]. Mechanistic Diamond Princess analyses likewise separate route attribution and environmental states from individual operational response [20]. Indoor-air reviews show why a physically detailed airflow layer matters for respiratory pathogens, especially when aerosols, recirculation, filtration, and room-to-room transport are the intervention targets [148–150]. CTB retains that branch for respiratory validation and HVAC engineering studies, while using the native solver when the tested object is surveillance, isolation, compliance, or VSP-style escalation rather than duct topology.

6.2 Paired-run validation design

The validation used paired simulations: each native run was matched to a CONTAM run with the same vessel platform, pathogen profile, initial seed, voyage duration, response configuration, and random-number stream.

Pairing makes the test stronger than comparing aggregate means because any solver-induced change should appear as a per-seed divergence in monitored aggregate state counts or voyage-level outcomes.

For each epoch, the comparison recorded susceptible, exposed, infected, symptomatic, recovered, immune, quarantined, isolated, and new-infection counts, then compared derived outcomes at voyage end.

The derived metric set included attack rate, peak prevalence, outbreak occurrence, peak epoch, detection epoch, detection lag, effective reproduction proxy, quarantine-person-epochs, and final susceptible fraction.

The paired design also protects against a common validation error in stochastic agent-based models. Two solvers can produce indistinguishable campaign means while disagreeing on individual seeds, especially near fade-out thresholds. Conversely, two solvers can disagree in environmental mass while preserving every infection event if the exposure calculation remains on the same side of the dose-response draw for each susceptible agent. The validation therefore used the agent state as the primary endpoint and environmental mass as a secondary diagnostic. This is consistent with calibration practice in dose-response models, where the fitted exposure scale and the endpoint definition must be interpreted together rather than as independent physical measurements [59–61].

6.3 Aggregate state-count agreement

The transport-solver validation found zero aggregate-level deltas in 200 paired runs.

The comparison yielded 6,720 matched state-count cells: 200 paired runs times 4 monitored epoch checkpoints times 8 monitored state fields, plus 320 matched voyage-level summary checks. It was not an hourly agent-history comparison.

No paired run changed attack rate, peak prevalence, outbreak occurrence, peak timing, detection timing, detection lag, quarantine burden, or final susceptible fraction.

The 200 pairs covered norovirus, influenza, measles, and SARS-CoV-2, so the result is not a single-pathogen smoke test.

It is a solver-equivalence result for the aggregate outcome variables used in the present campaign analyses, not proof that individual-indexed infection, quarantine, or isolation histories were identical.

The finding supports use of the native analytical solver for campaign-scale intervention com-

parisons. For the calibrated norovirus setting, the conclusion is especially direct: the observed endpoint is governed by dose-response, contact and environmental route weighting, shedding, and the VSP response backstop, not by small differences between native and CONTAM environmental pools. For respiratory pathogens, the result is narrower. It shows that the current campaign parameterization is not sensitive to the tested solver differences, but it does not eliminate the need for CONTAM when the experiment perturbs ventilation, filtration, recirculation, cabin leakage, or isolation-room placement. Cruise-ship respiratory studies indicate that those mechanisms can affect route attribution and intervention ranking [21, 42, 70, 151].

6.4 Environmental mass discrepancy

The two solvers did not produce identical environmental mass.

The broader validation notes report median CONTAM/native pathogen-mass ratios of approximately $1.67\times$ for norovirus, $1.42\times$ for SARS-CoV-2, $1.41\times$ for influenza, and $1.28\times$ for measles.

A regenerated norovirus quick-pair check gave a median mass ratio of $1.57\times$.

Some per-epoch ratios are much larger when the native denominator is close to zero, so median ratios are the interpretable diagnostic.

The discrepancy is meaningful for concentration-sensitive observables, but it did not propagate into the aggregate agent outcomes measured in the paired validation.

Three mechanisms explain this separation between environmental mass and epidemiological state. First, dose-response saturation can make a $1.3\text{--}1.7\times$ mass change epidemiologically silent when susceptible agents are already on the high-probability part of the infection curve. Second, CTB applies pathogen-specific route weights after environmental transport, so total zone mass is not a one-to-one map to infection probability. Third, shipwide mass can differ in zones or epochs that contribute little to occupied-agent dose. A mass discrepancy is therefore a transport diagnostic, not automatically an outbreak-outcome discrepancy. This interpretation is consistent with the Diamond Princess literature, where aerosol, fomite, short-range, and long-range contributions are inferred through coupled assumptions rather than observed as a single directly measured state [20, 71].

The $1.28\times$ to $3.17\times$ across-pathogen and across-zone median mass-ratio range between native and CONTAM solvers ($1.28\times$ for measles at the lower bound and higher zone-specific norovirus medians at the upper bound) translates to 0.11 to 0.50 $\log_{10}$ units. For qPCR-based environmental surveillance, this corresponds to 0.36 to 1.66 Ct values, assuming 3.32 Ct per $\log_{10}$ decade. Typical RT-qPCR coefficient of variation at the limit of detection is $\pm 1\text{--}2$ Ct, and cruise wastewater surveillance studies report detection of norovirus GII in 100% of untreated sewage samples at concentrations well above the limit of detection [152]. Under the single-Ct-per-log conversion used here, the solver mass discrepancy falls within the same order of magnitude as reported qPCR variability

at the limit of detection; a stronger noise-floor claim would require paired native-CONTAM environmental sampling and per-assay recovery data. Separately, air-sampler and wastewater surveillance operate on different matrices (aerosol vs. liquid), so the airborne-solver discrepancy is not by itself an uncertainty bound on wastewater qPCR. Environmental surveillance stoplights calibrated to one solver would nonetheless require an approximately $0.2 \log_{10}$ adjustment under the other; this indicates the solver-agnostic conclusion for wastewater is limited to whether the assay is above or below the detection threshold at typical outbreak concentrations, not to precise concentration estimation.

6.5 Implications for surveillance calibration

The agent-equivalence result should not be used to transfer environmental surveillance thresholds without recalibration.

Air samplers, surface swabs, and wastewater assays respond to concentration, sampling location, shedding into environmental reservoirs, assay recovery, and reporting thresholds.

Those observables are closer to the environmental mass ledger than to final attack rate.

If CONTAM produces a persistent $1.28\text{--}3.17\times$ shift in airborne or deposited mass, then a threshold calibrated under the native solver can fire at a different epoch under CONTAM even when the aggregate infection history is unchanged.

Environmental surveillance studies on ships and congregate systems show the same issue: detection is a measurement model with recovery efficiency, target choice, temporal smoothing, and threshold selection, not a direct census of infections [32, 58, 77, 152].

For this monograph, the operational rule is therefore solver-specific calibration for environmental instruments and solver-agnostic interpretation for the validated agent outcomes. Native-solver campaign outputs can be used for attack-rate, quarantine-burden, detection-lag, and economic comparisons within the validated parameter range. Environmental thresholds, air-sampler Ct cutoffs, zone prioritization, and HVAC intervention studies should be calibrated or rerun under CONTAM. This rule keeps the fast solver available for factorial campaigns while preserving the physically detailed branch for the cases where airflow is the object of inference.

6.6 When CONTAM matters

CONTAM matters when the scientific question is about airflow rather than about the response cascade.

Examples include respiratory-pathogen validation, recirculation versus outdoor-air scenarios, filtration upgrades, isolation-room placement, passenger-cabin leakage, and zone-specific air-sampling design.

It also matters when a claim depends on absolute environmental concentration rather than on internally calibrated infection outcomes.

The paired validation does not make CONTAM redundant; it defines the boundary where the native solver is sufficient.

Within that boundary, 200 paired runs and 6,720 matched monitored state-count or summary cells justify native transport for campaign-scale intervention sweeps.

Outside that boundary, especially for respiratory pathogens and HVAC interventions, CTB should use the CONTAM branch or repeat the paired validation after the airflow intervention is introduced.

Chapter 7

Detection, Intervention, and Pre-Boarding Screening

7.1 Scientific claim

Mid-voyage response saturates early, while pre-boarding screening has a larger decision value because it acts before infectious passengers or crew enter the shipboard contact network. In C14 and C14b, syndromic or cascade response reduced final attack rates for all modeled pathogens, and wearables, wastewater, and multiplex testing added less than one percentage point for most controlled comparisons. The boundary decision model changes the question from “which signal detects an outbreak first” to “which pre-boarding policy reduces the expected cost of a voyage.” On that criterion, incentivized wearable adoption followed by confirmatory testing is the preferred active policy across most prevalence, pathogen, and platform cells. The boundary model is anchored to pathogen-specific introduction prevalence rather than a generic boarding risk, because norovirus, SARS-CoV-2, influenza, and measles differ in community incidence, asymptomatic fraction, presymptomatic infectiousness, seasonality, and surveillance observability [92, 102, 103, 153–155].

7.2 Mid-voyage surveillance results: response saturation

The active-outbreak campaigns compared **none_true** against syndromic response, cascade response, cascade plus multiplex testing, wearable variants, and wastewater surveillance. Table 7.1 consolidates the cross-pathogen result. The uncontrolled attack rate is the mean attack rate under **none_true**. The interventions in the comparison correspond to the operational families described in the cruise literature: syndromic case recognition, isolation, intensified sanitation, diagnostic testing, quarantine, wastewater surveillance, wearable or symptom-based monitoring, and ventilation or filtration adjustments [5, 6, 24, 32, 42, 77, 80, 152]. The controlled range reports the span across syndromic and advanced response strategies in the C14/C14b intervention summaries.

Table 7.1: Cross-pathogen intervention comparison for active mid-voyage response. Attack rates are percentages. The advanced marginal gain is the best advanced strategy relative to syndromic response; negative values indicate that syndromic response was already the lowest mean in the supplied summary. Values are aggregated C14/C14b point estimates; underlying replicate counts are reported in Appendix A.4, and the supplied intervention summary did not include confidence intervals for the strategy differences.

Pathogen	Uncontrolled AR	Syndromic AR	Controlled range	Best advanced gain	Interpretation
Norovirus	17.21	5.87	5.23–5.87	0.64	Advanced signals add < 1 pp
SARS-CoV-2	20.72	0.38	0.49–0.86	-0.11	Response almost eliminates onboard spread
Measles	37.91	1.32	1.15–2.17	0.17	High immunity leaves little preventable residual
Influenza	66.08	1.89	1.20–2.14	0.69	Fast control after first trigger
<i>C. difficile</i>	12.03	0.18	0.18–0.55	0.00	Clinical response dominates tested add-ons

Norovirus is the limiting enteric case. The uncontrolled mean was 17.21%, syndromic response reduced it to 5.87%, and the best advanced mean in the supplied comparison was 5.23%. That 0.64 percentage-point difference is small compared with the 11.34 percentage-point reduction obtained by activating response at all. Because the supplied summary reports means without run-level replicate confidence intervals, the “< 1 pp” saturation statement is a point-estimate description consistent with response saturation, not a formal test that the true advanced-vs-syndromic gap is bounded below one percentage point. For SARS-CoV-2, the analogous contrast is 20.72% to 0.38% under syndromic response, with advanced strategies spanning 0.49–0.86% in this summary; by the table convention, the best advanced marginal gain is therefore -0.11 percentage points because syndromic response is already the lowest mean. Influenza begins at a much higher uncontrolled attack rate, 66.08%, but falls to 1.20–2.14% after response.

This saturation result should be interpreted operationally rather than as a negative result for wearable or wastewater instruments. Once the voyage has entered an active response state, intensified isolation, sanitation, and case management remove much of the remaining preventable transmission. Additional signals then compete for a small residual benefit. That pattern is consistent with the cruise-ship norovirus literature, where the timing of reporting and response is a major determinant of outbreak size [5, 5, 9, 10, 10, 16, 91]. It is also consistent with respiratory cruise outbreaks, where late detection after shipboard amplification can leave few low-cost options [18, 19, 19, 23, 23, 24].

7.3 Why boundary surveillance matters more than mid-voyage surveillance

Boundary surveillance and mid-voyage surveillance are evaluated on different endpoints and comparators in this monograph: mid-voyage strategies are compared incrementally to an active syndromic baseline on attack rate, while pre-boarding policies are compared with no screening on an expected voyage-cost bundle. The following argument is therefore about the mechanism through which each acts on outcomes, not a like-for-like comparison of two evaluations. Boundary surveillance acts on introductions rather than on transmission that has already occurred. The relevant empirical object is the probability that at least one infectious traveler boards, which scales as $1 - (1 - \pi_{inf})^N$ and therefore becomes large even when individual prevalence is below 0.1% on large ships [92, 102, 153–156]. In a mid-voyage setting, a wearable flag, wastewater signal, or multiplex result can only change the course of an outbreak after at least one infectious person has boarded, generated contacts, and possibly triggered secondary cases. At pre-boarding, the same signal can reduce the number of infectious people who board. The relevant state variable is therefore K_{board} , the number of infectious passengers and crew after screening, rather than the time of the first onboard alert.

The economic objective also changes at the boundary. For a cruise operator, the value of screening is not only the number of cases prevented. It includes avoidance of Vessel Sanitation Program escalation, voyage disruption, regulatory reporting, cleaning surges, denied-boarding compensation, and reputational loss. These cost categories are deliberately operational rather than biological: published VSP and COVID-era cruise analyses show that outbreak management invokes public reporting, enhanced cleaning, consultation or investigation, passenger communication, travel restrictions, and sometimes itinerary or disembarkation changes [5, 16, 24, 132, 157]. The decision model scores a policy by expected voyage cost, then reports the value of information per passenger as

$$\text{VoI}_{\text{pax}} = \frac{E[C_{P0}] - E[C_p]}{N_{\text{pax}}}, \quad (7.1)$$

where $P0$ is no screening, p is the screening policy, and N_{pax} is the passenger capacity for the platform. Positive values mean the policy reduces expected cost relative to no screening.

7.4 Embarkation prevalence anchors for π_{inf}

The decision model uses π_{inf} as the probability that a randomly selected boarding passenger or crew member is capable of transmitting at embarkation. This is narrower than PCR positivity, seropositivity, or wastewater signal. A useful approximation is incidence multiplied by infectious duration, adjusted for asymptomatic and presymptomatic infection and for under-ascertainment. The ranges below are scenario anchors for the x-axis, not fixed biological constants.

Norovirus illustrates why infectious prevalence and marker prevalence must be separated. Hall,

Table 7.2: Pre-boarding infectious-prevalence anchors used to interpret the decision grid. Ranges are probabilities that a randomly selected embarking person is infectious at boarding.

Pathogen	Low	Moderate	High	Main evidence and interpretation
Norovirus	0.01–0.05%	0.05–0.20%	0.20–0.80%	Incidence estimates of 19–21 million annual U.S. illnesses, winter concentration, challenge shedding, and asymptomatic PCR meta-analyses support a narrow infectious-prevalence band; stool-RNA positivity is a separate sensitivity quantity and can be much higher [51, 53, 92, 134, 158].
SARS-CoV-2	0.05–0.30%	0.30–1.0%	1.0–3.0%	Probability surveys are no longer continuous, so CDC/Biobot wastewater trends and RESP-NET are used as relative scalars; presymptomatic and asymptomatic transmission makes symptom-only screening insufficient [54, 87, 153, 156, 159–161].
Influenza A	0.005–0.05%	0.05–0.30%	0.30–1.0%	FluView and FluNet provide virologic and syndromic indicators rather than point prevalence, so the band is an incidence-duration conversion with strong temperate seasonality and possible year-round tropical importation [27, 28, 55, 154, 155].
Measles	0–0.00001%	0.00001–0.001%	0.001–0.02%	Measles is a rare, clustered importation risk; two-dose MMR is approximately 97% effective, and high scenarios apply to defined outbreaks, undervaccinated communities, or exposed travel groups rather than to ordinary national averages [4, 29, 102, 103].

Lopman, and colleagues provide the incidence anchor; Ahmed et al. show that norovirus is a major etiology among gastroenteritis cases; Qi et al. estimate asymptomatic stool positivity near 7% globally, with higher values in outbreak contexts; and challenge or outbreak studies show prolonged RNA shedding [51, 53, 92, 134, 158]. A 7% stool-positive sensitivity analysis is useful for stress testing wastewater or PCR detection, but it should not be used as the default probability of high-transmission infection at boarding.

For SARS-CoV-2, the practical surveillance stack is local wastewater trajectory, hospitalization lag, and recent probability-survey context. Wastewater has lead-time value but cannot be converted to a unique percent infected without assumptions about variant-specific shedding, sewer flow, recovery, and infection-age distribution [78, 153, 159, 161, 162]. Pre-boarding symptom checks also miss a material fraction of risk because viral load often peaks near symptom onset and transmission can occur before symptoms [54, 87, 163].

Influenza and measles require different boundary logic. Influenza prevalence should follow port-and-origin FluView or FluNet activity and the passenger origin mix, because a globally sourced cruise can import influenza when the embarkation country is out of season [55, 154, 155]. Measles should be modeled as a rare-event, exposure-and-immunity problem. For ordinary source catchments, π_{inf} is close to zero; for a known contact, affected community, or crew cohort with uncertain documentation, the population prevalence band should be replaced by individual exclusion, post-exposure prophylaxis, and immunity verification logic [29, 102, 103].

Published embarkation-screening performance data for cruise lines remain sparse, although recent cruise-health work has specifically evaluated passenger preferences for onboard communicable-disease screening technologies and port-preparedness requirements for passenger ships [164, 165].

Evidence from thermal-screening reviews and airport programs shows low sensitivity for asymptomatic, presymptomatic, afebrile, and antipyretic-masked infection, while cruise outbreak reports show that symptom-based triage alone could not prevent SARS-CoV-2 or norovirus amplification [4, 23, 86, 166]. The model therefore treats questionnaires and temperature checks as low-cost deferral and triage layers, not as clearance tests.

7.5 Pre-boarding decision model

The Monte Carlo decision model combines four inputs: pre-boarding prevalence, wearable adoption and accuracy, confirmatory testing, and a posterior outbreak surface indexed by the number of infectious individuals who board. Each scenario draws infectious status for passengers and crew, applies the screening policy, looks up the corresponding outbreak outcome at K_{board} , and computes expected cost. This structure mirrors decision analyses in which testing, tracing, isolation, and occupancy controls alter the number of infectious contacts entering or remaining in the modeled network [24, 46, 47]. The corrected analysis contains 528 scenarios per pathogen: four platform classes, twenty-two prevalence grid entries, and six policies. Each scenario uses 2,000 Monte Carlo voyages.

The six policies are:

P0 No pre-boarding screening.

P1 Advisory-only screening. In the implemented model, this has no operational screening charge and does not change boarding, so it is a same-path baseline for P0.

P2 Bring-your-own-device wearable adoption with confirmatory testing; boarding is denied only after a positive confirmatory test.

P3 Wearable flag alone; boarding is denied after an anomaly without confirmatory testing.

P4 Incentivized wearable adoption, with at least 70% passenger adoption, followed by confirmatory testing.

P5 Crew-mandatory adoption with the same confirmatory-test boarding rule as P2.

Adoption and coverage assumptions used in the runs: P2 assumes voluntary BYOD adoption at 35% among passengers and 40% among crew; P4 assumes at least 70% passenger adoption after incentives and 60% crew adoption; P5 assumes 95% crew adoption enforced through operational policy. Confirmatory-test turnaround is assumed to fit within a same-day pre-boarding window; denied-boarding uptake is treated as full (no boarding of denied passengers). Compliance with the initial denial decision, appeals processing, and rebooking are not modeled and should be added when partner data become available.

Wearable sensitivity and specificity are pathogen-specific. The decision runs used sensitivity/specificity of 0.55/0.85 for norovirus, 0.65/0.85 for influenza, 0.70/0.85 for SARS-CoV-2, and 0.75/0.88 for measles. Confirmatory testing used sensitivity 0.90 and specificity 0.98. The value is a scenario parameter representing near-boarding molecular or high-quality rapid testing; real instruments vary by pathogen, specimen, timing, viral burden, and implementation [72–75, 167–169]. The mid-cost scenario assigned \$1 per wearable screen, \$75 per confirmatory test, \$2,000 per false-positive or true-positive denied boarding, \$400 per onboard case, \$500,000 per outbreak, \$2 million for VSP escalation, and \$2 million for reputational cost. The \$1 per-screen figure represents the per-passenger marginal cost of ingesting an existing wearable data stream (bring-your-own-device or an already-issued device) into the boarding decision layer, not the full retail cost of an incentivized wearable program; a scenario that internalizes device provisioning, incentives, and support at, e.g., \$50–\$150 per passenger narrows the positive P4 window on larger platforms and should be treated as a sensitivity target rather than an assumed baseline. These values are deliberately separable parameters rather than fixed biological assumptions. They should be updated with operator-specific compensation policies, port-health requirements, itinerary disruption costs, public-reporting rules, and passenger-demand effects; the peer-reviewed cruise literature supports the existence of these cost domains but not a universal dollar value for each [5, 24, 133, 157].

7.6 Outbreak surfaces from the k -sweep campaign

The boundary model uses k -indexed outbreak surfaces generated from 17,600 simulation runs. The surfaces are needed because confined-environment outbreaks are overdispersed: final size depends on introduction count, timing, cabin and venue exposure, crew roles, and intervention delay rather than on mean prevalence alone [23, 65–67, 91]. Each surface is keyed by platform class, pathogen, baseline response, and the number of infectious individuals who board. For each k , the surface stores the probability of VSP triggering, expected attack rate, probability of acceleration, expected onboard cost, expected peak epoch, and run count. The decision model interpolates these curves when the Monte Carlo draw produces a non-integer lookup point, although the implemented screening pipeline returns integer K_{board} .

The surfaces explain why screening value rolls over. When K_{board} is low, intercepting one infectious individual can move a voyage from a high-risk point on the surface to a low-risk point. When prevalence is high, the no-screening voyage is already near the high-risk plateau, and screening cannot intercept enough infectious people to move the operating point out of saturation. False positives and screening costs then continue to rise while marginal VSP avoidance falls. The rollover is therefore a decision-surface property, not a claim that prevalence is unimportant; at high prevalence the appropriate action may shift from individual screening to voyage-level modification, postponement, or a stronger mandate [24, 132]. This produces the large-ship rollover in value of information, where the “rollover” is defined as the prevalence at which incremental screening value first crosses zero and is distinct from the “peak” prevalence at which value of information is maximized; both

are reported separately in Table 7.3.

Surface shape differs by pathogen. Influenza reaches VSP-trigger probabilities above 0.5 at $k = 1$ and above 0.9 at $k = 2$ on all four platforms in the supplied surface. Norovirus reaches 0.5 at $k = 2$ and 0.9 at $k = 4$ or $k = 5$ depending on platform. Measles reaches 0.5 at $k = 3$ on expedition ships and $k = 5$ on the other classes, with the 0.9 threshold occurring only at higher k for several large-ship curves. SARS-CoV-2 is flatter on the large-ship surfaces in this response setting; among the supplied curves, only the expedition surface crosses 0.5.

7.7 Cross-pathogen value of information

Figure 7.1 shows the corrected cross-pathogen decision surfaces. The most consistent result is the platform gradient. Expedition ships have the highest value of information, and the value does not become negative over the tested prevalence grid for P4. The peak P4 value is \$4,914 per passenger for norovirus, \$6,072 for influenza, \$10,411 for SARS-CoV-2, and \$10,066 for measles on expedition ships. These peaks occur at higher prevalence on expedition ships than on larger platforms because a small absolute number of introductions can move the whole voyage across a high-cost response threshold.

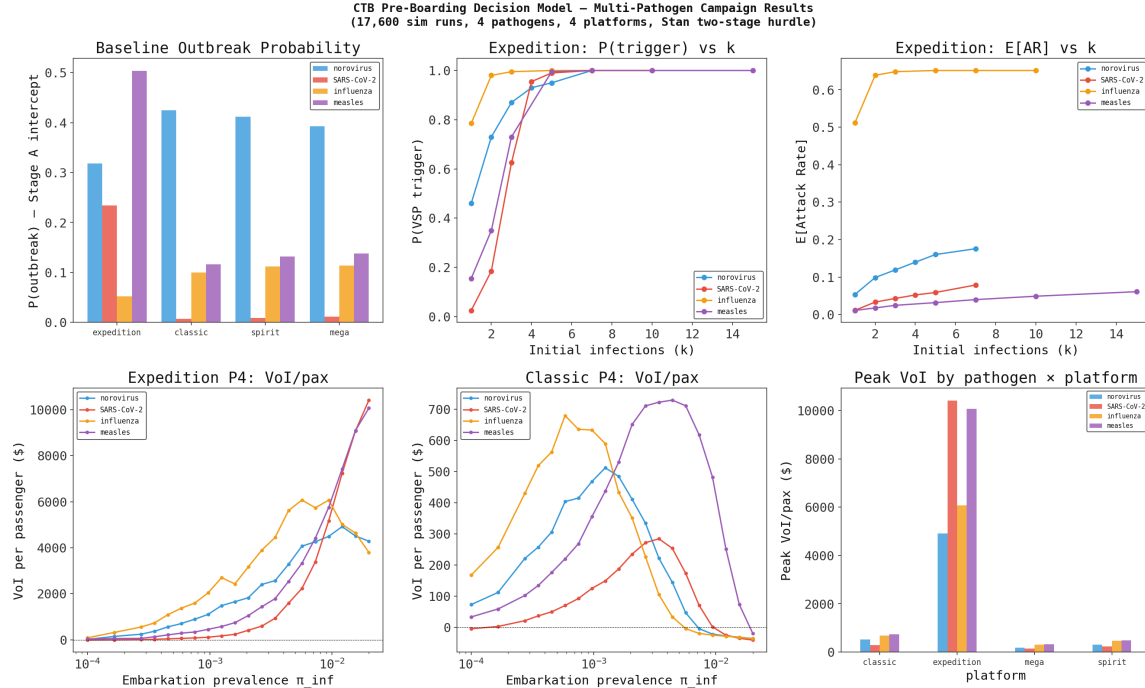


Figure 7.1: Corrected cross-pathogen value-of-information analysis for pre-boarding screening. Values compare screening policies against P0 no screening under the mid-cost scenario. P4, incentivized wearable adoption plus confirmatory testing, has the highest peak value and is the preferred active policy across most decision cells; large ships show rollover when outbreak probability saturates.

SARS-CoV-2 has the highest peak screening value despite lower VSP-trigger probability on

large platforms. Cruise COVID-19 experience gives this result external plausibility: rare large outbreaks had severe operational consequences even when many voyages had few cases [23–25, 166]. The reason is that the expedition curve remains highly sensitive to introductions across the tested prevalence range, while the cost bundle attached to a triggered voyage is large. At the 2% expedition peak, the MC calculation uses a mean of 4.07 introduced and 2.11 boarding infectious SARS-CoV-2 cases after screening, with expected trigger probability 0.335; the high value is therefore driven by avoided cost on a steep, non-saturated surface rather than by assuming five or more post-screening infectious boarders. Influenza has the narrowest positive-VoI window on large ships in the tested cost bundle: its outbreak surfaces saturate at very low k , so screening is valuable at low prevalence but loses value rapidly once multiple infectious people are likely to board. It is therefore not “ineffective” on large ships, but its useful prevalence range is narrow, which is why influenza screening on large platforms should be treated as a conditional low-prevalence tool rather than a routine layer. Measles has a wider screening window than influenza because the modeled population has high non-susceptibility, here 92%, so introductions do not saturate the response surface as quickly. The 92% non-susceptibility rate used in the model reflects the population-level effective immunity in a vaccinated cruise passenger cohort, which is lower than the 97% two-dose MMR vaccine efficacy due to waning immunity, single-dose recipients, and unvaccinated travelers. For measles, the operational equivalent of screening is primarily immunity verification and exposure exclusion, since presymptomatic infectiousness begins before rash and asymptomatic carriage is not the relevant reservoir [29, 102, 103, 169]. Norovirus is intermediate: it produces positive P4 values across the full expedition grid, but the large-ship positive window closes by approximately 0.27–0.57% prevalence depending on class.

Table 7.3: P4 value-of-information peaks under the mid-cost scenario. Prevalence is the pre-boarding infectious prevalence at the peak. Values are dollars per passenger relative to P0 no screening.

Pathogen	Expedition		Classic		Spirit		Mega	
	Peak VoI	Prev.	Peak VoI	Prev.	Peak VoI	Prev.	Peak VoI	Prev.
Norovirus	4914	1.207%	512	0.125%	298	0.097%	185	0.045%
SARS-CoV-2	10411	2.000%	284	0.342%	232	0.206%	148	0.125%
Influenza	6072	0.567%	679	0.058%	469	0.045%	297	0.027%
Measles	10066	2.000%	729	0.440%	479	0.266%	330	0.160%

P4 is not uniformly best in every row of the corrected grid. It is the top active policy in 277 of 352 pathogen–platform–prevalence entries. The remaining 75 entries are mostly high-prevalence large-ship endpoints where all active policies have negative value and the less intensive P1 policy, or in two entries P2, loses less money. This is not a recommendation to use P2 at high prevalence, nor does the VoI rollover imply that screening is inherently useless in high-prevalence regimes. The rollover occurs because introduction count K_{board} saturates the outbreak surface: when community prevalence is high enough that even aggressive screening cannot reduce K_{board} below the threshold at which $P(\text{trigger}) \approx 1$, individual-level interception no longer changes the voyage-level outcome.

This saturation point is an *operational decision boundary*. Below it, pre-boarding wearable screening is cost-effective. Above it, the correct intervention may shift from individual screening to voyage-level actions such as broader pre-departure testing, embarkation delay, port-of-call restrictions, or—in extreme cases—voyage cancellation. The VSP manual supports AGE reporting and sanitation escalation, while the IHR provides a general port-health framework; neither source is treated here as a universal pre-boarding-testing protocol. This decision-surface implication should be validated with dedicated voyage-level campaigns before being used to set policy [6, 7]. Operators who continue running individual screening programmes past the rollover are spending more on false positives and compensation than they save in avoided outbreaks. The rollover prevalence therefore deserves explicit reporting alongside VoI peak values, as it marks the transition between the screening regime and the population-level control regime.

7.8 Policy recommendations by platform class

Under the tested mid-cost scenario, expedition ships have positive P4 value of information across the tested prevalence grid and are the strongest candidate for treating pre-boarding screening as a standing risk-management option for all four pathogens evaluated here. The cost-sensitivity grid strengthens this recommendation for expedition vessels: P4 remains positive even under the low-cost bundle, and a \$100 per-passenger device-provisioning charge reduces peak value by only about \$95 per passenger. The recommendation remains conditional on the assumed pathogen-specific wearable performance and generic regulatory-disruption cost bundle. This recommendation reflects platform geometry and threshold sensitivity in the model, and it should be combined with the literature-supported baseline package of illness reporting, isolation, sanitation, vaccination verification where relevant, and port-health communication [4, 6, 7, 131]. Their P4 value remains positive across the tested prevalence grid, and the maximum value is measured in thousands of dollars per passenger. Small absolute changes in K_{board} have large consequences because the voyage has less capacity to absorb introductions before crossing a response threshold.

Classic and spirit-class ships should use P4 selectively when pre-boarding prevalence is in the positive window. For norovirus, the positive P4 window extends to about 0.57% on classic ships and 0.44% on spirit ships. For influenza, the window is narrower, ending near 0.44% on classic ships and 0.27% on spirit ships. For measles and SARS-CoV-2, the window is wider than influenza because the surfaces roll over later.

Mega ships require the most conservative screening trigger. The peak P4 value is positive for all four pathogens, but the positive window closes earliest: approximately 0.27% for norovirus, 0.34% for SARS-CoV-2, 0.16% for influenza, and 0.57% for measles. At higher prevalence, the relevant operational choice may shift from individual screening to voyage modification, targeted cancellation, pre-boarding delay, or a pre-departure testing mandate with a different cost structure. COVID-era maritime guidance and modeling reached the same qualitative conclusion: bundled controls,

occupancy decisions, voyage length, testing cadence, and public-health coordination determine whether individual-layer interventions remain worthwhile [24, 132, 157].

7.9 Cost and accuracy sensitivity

The cost-sensitivity grid varies the outbreak-cost bundle and explicit device provisioning while retaining the pathogen-specific wearable accuracy assumptions used in the corrected Monte Carlo analysis. Accuracy remains a model parameter: higher sensitivity reduces K_{board} when the outbreak surface is steep, and higher specificity reduces false-positive denied boarding, especially at low prevalence. The stored sensitivity runs therefore quantify the cost side directly and retain the accuracy interpretation from the policy comparison: P3 is penalized because it denies boarding on the wearable flag alone, while P2 and P4 use confirmatory testing to control false-positive cost.

Table 7.4: P4 expedition-class peak value of information under cost and device-provisioning sensitivity. Values are dollars per passenger relative to P0 no screening. Device cost is an added per-passenger provisioning or support charge.

Pathogen	Low/\$0	Low/\$100	Mid/\$0	Mid/\$100	High/\$0	High/\$100
Norovirus	\$1,200	\$1,106	\$4,914	\$4,820	\$29,542	\$29,447
SARS-CoV-2	\$2,544	\$2,449	\$10,411	\$10,316	\$62,584	\$62,489

Device provisioning at \$100 per passenger reduces peak P4 value by about \$95 per passenger in both pathogens. That decrement is small on expedition-class vessels because the avoided-response term dominates the marginal screening cost. Under the low-cost bundle, expedition screening remains positive, with peak values of \$1,200 per passenger for norovirus and \$2,544 per passenger for SARS-CoV-2 before the device charge. The same low-cost bundle is not sufficient on the largest platform when device provisioning is included: mega-class P4 never pays off under low cost plus a \$100 device charge, with peak value of information of -\$51 per passenger for norovirus and -\$60 per passenger for SARS-CoV-2.

The high-cost bundle expands the value scale rather than changing the mechanism. Expedition-class norovirus rises from \$4,914 per passenger in the mid-cost, no-device case to \$29,542 per passenger in the high-cost, no-device case, a six-fold increase. SARS-CoV-2 rises from \$10,411 to \$62,584 per passenger over the same comparison. Break-even prevalence shifts modestly with device cost and cost bundle but remains very low for expedition-class vessels: the SARS-CoV-2 expedition break-even point is 0.027% in the mid-cost, no-device case and 0.097% when the \$100 device charge is added; norovirus remains positive from the lowest tested grid point in the mid-cost, no-device case and crosses by 0.017% with the \$100 device charge. Large-ship rollover remains the limiting feature because screening loses value once the no-screening voyage is already near the high-risk plateau.

The current cost scenario sets the “regulatory disruption” bundle at \$4.5 million per triggered voyage when outbreak, VSP, and reputation terms are combined. VSP is an AGE-specific rule,

so this bundle is applied to non-AGE pathogens as a proxy for analogous regulatory-disruption events such as port-health investigations, quarantine orders under IHR (2005) Article 28, itinerary changes, and public-reporting cascades. The dollar value is not literally a VSP charge for SARS-CoV-2, influenza, or measles; it is a separable cost parameter that should be replaced with operator- and pathogen-specific cost data when available. Wastewater surveillance should be analyzed as a complementary early-warning layer rather than a direct boarding-denial test, because ship and community studies show detection feasibility but still require sampling design, normalization, and action thresholds [32, 77, 78, 152, 170–172].

7.10 Implication for intervention design

The intervention programme should be split into two layers. Mid-voyage surveillance should be retained as a response-quality and situational-awareness layer, with expectations calibrated to the saturation result in Table 7.1. Within that layer, contact tracing and isolation improve response quality, ventilation and HEPA/outdoor-air interventions should be scenario-tested where respiratory routes are active, wastewater can provide ship-level situational awareness, and vaccination or immunity verification is pathogen-specific rather than a generic screening substitute [20, 29, 41, 42, 47, 152]. Pre-boarding screening should be treated as the higher-value decision layer because it moves the voyage along the K_{board} outbreak surface before amplification begins. For the tested parameterization, P4 is the preferred operational starting point: broad incentivized adoption, confirmatory testing before denied boarding, and platform-specific prevalence thresholds that prevent screening from operating after value has rolled over. The recommendation should be revisited when operator-specific costs, passenger origin data, vaccine requirements, instrument pricing, and prevalence nowcasts are available, because those inputs control the break-even thresholds more strongly than the generic biological parameters alone [102, 103, 153–155, 159].

Chapter 8

Spatial-Topology Generalization Stress Test

To probe whether the transmission engine’s behavior remains coherent when the spatial graph is not a cruise deck plan, we instantiated the model on two non-terrestrial topologies drawn from the *Star Trek* franchise. This chapter is a consistency check on the model’s spatial abstraction: passing it does not demonstrate topology invariance across all possible spatial graphs, but failing it would indicate cruise-specific hardcoding.

8.1 Scientific claim

The Starfleet campaign is a deliberately playful stress test of CTB, but it is not a toy model in the technical sense. The same agent state machine, dose accumulation, route-weighted transmission, stochastic introduction process, and response logic used for cruise ships can be instantiated on a different confined environment by replacing the spatial topology and schedule layer. In 5,520 starship runs, CTB reproduced the same qualitative pattern seen in Chapters 5 and 7: uncontrolled pathogens often saturate the population, while early operational response can reduce final attack rates by large margins. The result supports the intended scope of CTB as a confined-environment outbreak model rather than a cruise-only simulator [5, 6, 10, 64].

8.2 From cruise decks to starship decks

The Starfleet platforms replace passenger cabins, dining rooms, theaters, galleys, crew spaces, and medical centers with bridge, engineering, sickbay, transporter rooms, crew quarters, corridors, and mess spaces. This is a change in topology, not in pathogen mathematics. Contacts still arise from co-location, role schedules, shared air or surface exposure, and stochastic encounter structure. That distinction matters because published shipboard and confined-environment models often disagree

less about infection biology than about who shares space with whom, when, and under what response rule [24, 46, 47].

Two canonical hulls were used. The Constitution-class implementation represents a smaller Enterprise-scale crew of 430 agents. The Galaxy-class implementation represents a larger Enterprise-D-scale complement; the archived campaign table contains 1,014 model agents, within two persons of the 1,012 planning value. No result below depends on those two agents. The important contrast is the same one used for the cruise platform ladder: population size, compartmentalization, and venue mixing vary while the pathogen and response code remain fixed.

8.3 Campaign design

The campaign contains 5,520 runs across four Starfleet pathogen labels: influenza, Psi 2000, Rigelian fever, and Barclay’s Protomorphosis Syndrome. The labels are fictional where appropriate, but the simulation objects are ordinary transmission profiles: introduction, susceptibility, dose scaling, shedding, route exposure, and response. Surveillance states included `none_true`, sickbay response, Starfleet Medical response, and, for the Galaxy-class respiratory/physiologic setting, combadge bioscan. This keeps the fun in the noun layer while keeping the epidemiology in the verb layer.

The comparison is intentionally severe. Under `none_true`, every Starfleet pathogen-platform cell reached a 100% outbreak fraction. Attack rates were also at or near saturation: Constitution-class influenza, Psi 2000, and Rigelian fever reached 1.00 mean attack rate, and Galaxy-class Barclay’s syndrome reached 1.00. Galaxy-class influenza under 30% immunity still produced a mean attack rate near 0.70, while the no-immunity condition was approximately 1.00. These values are the starship analogue of the uncontrolled cruise surfaces in Chapter 5: when a pathogen has enough contact opportunity before response, the final size becomes a poor discriminator of mechanism [65–67].

8.4 Response reduces final size, but topology decides how much room response has

Sickbay response changed the outcome substantially. On the Constitution-class platform, active response reduced mean attack rate from 1.00 to 0.34 for influenza, 0.28 for Psi 2000, and 0.26 for Rigelian fever. Outbreak fraction remained 1.00 for influenza, but fell below 0.30 for Psi 2000 and Rigelian fever. This separation between outbreak occurrence and conditional final size is the same interpretive problem as the VSP threshold in Chapter 5: triggering and final burden are different endpoints.

On the Galaxy-class platform, active response also reduced burden, but the larger contact system retained more residual spread. Barclay’s syndrome fell from 1.00 to about 0.51 mean attack rate

across active response modes, and its outbreak fraction fell from 1.00 to about 0.52. Galaxy-class influenza fell from a `none_true` mean of 0.85 across immunity strata to about 0.41 under active response, with outbreak fraction still 1.00. The same qualitative pattern appears in Chapter 7, where syndromic response cuts final attack rates sharply but leaves pathogen-specific residuals once the response state is active [17, 18, 23].

The smaller Constitution-class ship often contained better under response than the larger Galaxy-class ship. That does not mean small confined environments are always safer. In Chapters 5 and 7, small cruise platforms can also have high value of information because a few introductions can move the whole voyage across a costly threshold. The Starfleet result is narrower: after response activates, a smaller, more compartmentalized topology can leave fewer parallel transmission paths open, so sickbay can convert a universal exposure threat into a partial outbreak more often.

8.5 Contact topology is consistent with the Starfleet result

The Constitution-vs-Galaxy comparison confounds topology with population size, response capacity, and role structure, so the observed difference is consistent with topology mattering but does not isolate it from those covariates. The difference is easiest to see in venue logic. Bridge, engineering, and sickbay are high-importance nodes, but they are role-structured. A bridge officer, engineer, and clinician do not mix with the entire complement at each epoch unless schedules or alarms route them together. Open mess halls, large crew lounges, and broad corridor traffic behave more like cruise dining and entertainment venues: they create repeated weak ties across otherwise separate social clusters. CTB therefore generalizes by preserving the distinction between local intense contacts and population-bridging venues, rather than by assuming that all confined environments share the same contact matrix [20, 21, 27, 28].

This topology statement is also why the Trek campaign should not be read as a claim about Star Trek canon. It is a model identifiability check. If the code only worked on cruise ships, then replacing the deck plan with a starship would either fail technically or erase the platform gradient. Instead, the model produces plausible confined-environment behavior: uncontrolled saturation, response-mediated compression, persistent dependence on population topology, and different roles for outbreak probability and attack rate.

8.6 Interpretation

The Starfleet campaign is therefore a useful falsification exercise. CTB did not require a cruise-specific exception to simulate Constitution-class and Galaxy-class environments. It required only a new spatial graph and schedule grammar. The main cruise conclusion survives translation: platform architecture is epidemiologically visible before and after response, but observed attack rates are response-shadowed outcomes. In plainer terms, Starfleet Medical cannot repeal contact topology;

it can only act quickly enough that topology has less time to matter.

Chapter 9

Discussion

9.1 Intervention-shadowed observations

The central finding of this monograph is that cruise-ship outbreak data cannot be interpreted as passive samples from uncontrolled epidemics. The Vessel Sanitation Program (VSP) record is generated by a coupled process: illness must occur, the passenger or crew member must present or be counted, the case must meet a syndromic definition, the numerator must approach or cross a passenger- or crew-denominator-specific threshold, and the ship and public-health system must respond. The resulting endpoint is a post-response measurement, not an intrinsic attack rate of a pathogen on a vessel class [5, 5, 6, 50]. This is why the monograph treats observation and intervention as inseparable for calibration, and why the model architecture makes the observation engine and protocol engine part of the epidemic system rather than downstream reporting utilities.

The norovirus calibration makes the problem concrete. At $d_{\text{adj}}=10.6$, the no-response condition retained a large physical platform gradient: 23.4% attack rate on expedition, 22.6% on classic, 12.2% on spirit, and 10.6% on mega. Under syndromic detection with a VSP-like response, the same biological setting compressed to 6.7%, 7.1%, 4.6%, and 5.5%. The threshold sweep in Chapter 5 showed the same mechanism more directly: no response produced a 10.6–23.4% range; sick call alone produced 7.3–13.2%; VSP thresholds of 5%, 3%, and 1% compressed the ranges to 5.8–7.9%, 5.1–7.0%, and 4.3–6.0%. This gradient is a model output under the calibrated parameterization; no empirical uncontrolled cruise attack rates exist to constrain it directly. The identifiability experiment (Section 5.9) shows that alternative parameterizations along the dose–alpha ridge could reproduce the same post-response band with a different uncontrolled gradient. These values are not compatible with a calibration story in which ship architecture is irrelevant. They are compatible with a response system that truncates observed final sizes after detection.

The literature supports this interpretation. VSP public postings and MIDRS surveillance are threshold-selected, clinic-ascertained, jurisdictionally censored syndromic data; they exclude untreated illness, many below-threshold voyages, events outside the reporting window, and some

post-disembarkation disease [5, 5, 10]. The 2% and 3% AGE levels also activate monitoring, consultation, specimen collection, case restriction, intensified cleaning, and public communication rather than merely labeling an already completed event [6, 13, 16]. A model fitted to VSP records as uncontrolled final sizes would therefore underestimate transmission potential in settings where the response would have acted early. The correct calibration object is the distribution produced by exposure, symptoms, reporting, threshold-triggered response, and post-response final attack rate.

The VSP degradation sweep adds a direct robustness check. Across 6,360 runs, the response shadow remained intact under realistic operational variation. In the threshold–compliance panel, the platform gradient re-emerged only at a 20% VSP threshold with isolation compliance equal to 10%; the broader worst-case panel contained nine broken cells. When it did, the direction reversed: mega-class ships showed higher attack rates than expedition-class ships because dense contact geometry on large vessels overwhelmed weak response. This reversal validates the model’s spatial fidelity. The platform gradient is not an artifact of population size alone; it reflects the interaction between contact topology and response effectiveness.

9.2 Surveillance saturation within the voyage

The intervention campaigns show that once an active mid-voyage response exists, the marginal attack-rate benefit of additional surveillance modalities is small. In Table 7.1, norovirus fell from 17.21% under `none_true` to 5.87% under syndromic response; the best advanced strategy in the supplied summary reached 5.23%, a 0.64 percentage-point gain over syndromic response. SARS-CoV-2 fell from 20.72% to 0.38% under syndromic response, with advanced strategies in that comparison no better than the syndromic mean. Measles fell from 37.91% to 1.32%, with a best advanced gain of 0.17 percentage points, and influenza fell from 66.08% to a controlled range of 1.20–2.14%. Across these controlled comparisons, wearables, wastewater, multiplex testing, and cascade refinements generally added less than one percentage point after response was active.

This result should not be read as evidence that wastewater, wearable, or multiplex instruments have no operational value. It says that final attack rate becomes an insensitive endpoint after syndromic recognition has already activated isolation, sanitation, diagnostic testing, and case management. Advanced signals may improve time to detection, spatial attribution, false-alarm control, confidence in etiology, and the appropriateness of escalation. In the C14/C14b design, advanced instruments were evaluated inside the same voyage as an already active syndromic response rather than as pre-syndromic detectors; the residual attack-rate gap therefore describes marginal value after response activation, not the total value that an advanced signal acting before syndromic response could deliver. They may also matter more under weaker baseline response, different costs, lower syndromic reporting, or pathogens whose symptoms are nonspecific. The empirical literature supports retaining these instruments as imperfect but useful observers: wastewater can detect pathogen nucleic acid before clinical counts are complete, wearables can identify physiological anomalies for

respiratory infection, and diagnostic tests can sharpen action thresholds, but all are limited by sensitivity, specificity, adherence, sampling design, and reporting delay [77, 80, 81, 152].

The saturation finding is also a warning about evaluation design. A campaign that compares advanced instruments only after a strong syndromic response has fired is estimating residual marginal value, not the total value of information. The large gains in Table 7.1 came from activating response at all. The small differences among controlled strategies then reflect the remaining preventable transmission after the ship has already changed behavior. Future validation should therefore report endpoints that are closer to instrument function: lead time, avoided unnecessary escalation, isolation-days, confirmatory-test burden, confidence in route attribution, and the probability of crossing VSP or company thresholds.

9.3 Boundary surveillance and value of information

Pre-boarding screening has a different causal position. Mid-voyage surveillance observes an outbreak after at least one infectious person has entered the contact network. Boundary screening changes the number of infectious people who board, denoted in Chapter 3 as K_{board} . Because the probability of at least one infectious boarder is $1 - (1 - \pi_{inf})^N$, large ships can face near-certain introductions at individual infectious prevalences below 0.1%, while smaller ships often remain in a range where intercepting one case changes the voyage state. The prevalence anchors in Table 7.2 therefore matter: infectious prevalence is not PCR positivity, wastewater signal, seropositivity, or convalescent shedding [92, 102, 153–155, 158].

The corrected decision model found that incentivized wearable adoption followed by confirmatory testing (P4) was the preferred active policy in 277 of 352 pathogen–platform–prevalence entries. The high-value cases were concentrated on expedition ships, where peak P4 value of information reached \$4,914 per passenger for norovirus, \$6,072 for influenza, \$10,411 for SARS-CoV-2, and \$10,066 for measles (Table 7.3). Those values fall in the \$5K–\$10K per-passenger range for the highest-value expedition scenarios and remain positive across the tested expedition prevalence grid. On larger ships, P4 still had positive peaks, but the positive window was narrow: mega-class peaks occurred at 0.045% prevalence for norovirus, 0.125% for SARS-CoV-2, 0.027% for influenza, and 0.160% for measles. Once prevalence is high enough that multiple infectious people are likely to board, the surface saturates and individual screening no longer moves the voyage away from the high-cost plateau. “Peak” prevalence in Table 7.3 refers to the prevalence at which value of information is maximized; “rollover” or “break-even” prevalence refers to the higher prevalence at which value of information first crosses zero. These are separate quantities.

This rollover is consistent with, rather than uniquely explanatory of, the platform gradient. Under the tested cost bundle, Smaller ships benefit more from screening because there are fewer introductions to intercept and each intercepted case represents a larger shift in the introduction distribution. Large ships benefit when prevalence is low enough that one or two intercepted cases can

change the probability of triggering response, but the same large denominator makes introduction saturation occur quickly. At high prevalence, the decision should shift from voluntary screening to stronger pre-departure testing, boarding delay, itinerary modification, passenger communication, targeted cancellation, or other voyage-level actions. The economic result is therefore not simply that screening is good or bad. It identifies where a screening policy can still change the outbreak surface.

Pathogen ordering is platform-specific because the k -sweep surfaces differ. On expedition ships, SARS-CoV-2 had the highest peak P4 value, followed closely by measles, then influenza, then norovirus. This ordering reflects the cost of a triggered respiratory outbreak and the fact that the expedition SARS-CoV-2 surface remained sensitive across the tested prevalence range. On mega ships, the peak ordering was measles, influenza, norovirus, then SARS-CoV-2, with all peaks far below expedition values. Influenza saturated fastest on large ships, reaching high trigger probability at low k , while measles retained a wider screening window because the modeled population had high non-susceptibility. Norovirus was intermediate: positive across the full expedition grid, but rolling over on large ships around 0.27–0.57% prevalence depending on platform. These rankings should be revisited with operator-specific cost bundles, vaccination requirements, test pricing, passenger origin data, and live prevalence nowcasts [24, 86, 157].

9.4 Calibration, identifiability, and campaign scale

The calibration philosophy that follows from these results is restrictive. A single d_{adj} value can work across platforms only when the observation process includes VSP-like response. Norovirus at $d_{\text{adj}}=10.6$ is not a unique physical conversion from genome copies to infectious virions, and it is not a claim that platform effects disappear. It is an effective dose scale conditional on the shedding model, route weights, susceptibility assumptions, care-seeking process, VSP threshold, and response rules. The same distinction appears in the norovirus biology: symptoms, shedding, PCR positivity, and infectiousness are related but separable states; FUT2 secretor status and genotype affect susceptibility and severity; and GII.4 severity cannot be treated as a pure transmission multiplier [51, 59, 98, 99, 144].

The monograph therefore treats calibration as a joint fit to outbreak occurrence and conditional final size rather than as a fit to one attack-rate target. The boundary Stan analyses operationalize this by separating Stage A outbreak probability from Stage B conditional attack rate. For norovirus, Stage A contained 2,400 runs and 1,976 threshold-triggered events; Stage B fit the trigger-positive runs and showed lower conditional attack-rate intercepts for the larger spirit and mega platforms than for expedition and classic. For SARS-CoV-2, influenza, and measles, the boundary surfaces separated introduction sensitivity, response saturation, and immunity geometry. This two-stage hurdle structure is necessary because triggering and final size answer different questions. A parameter set can reproduce final attack rates while failing to reproduce outbreak probability, fade-out,

acceleration, or response burden.

The 77,310 simulation runs (69,750 core campaign-ledger runs plus 1,200 synthetic-recovery runs and 6,360 VSP degradation runs) are therefore a computational investment in inference, not only a larger scenario catalogue. Replicated stochastic runs make it possible to estimate outbreak probabilities, conditional final-size distributions, acceleration probabilities, trigger surfaces, and policy value with enough precision to fit and compare the two-stage statistical summaries. Small campaigns can illustrate mechanisms, but they cannot support the same separation of occurrence and severity, especially for rare introductions, high-dispersion processes, and policy rollovers. This is important for cruise ships because overdispersion is expected from biology, venue structure, crew roles, cabin sharing, environmental reservoirs, and intervention timing [65–68, 91].

Large simulation volume does not solve structural identifiability by itself. Structural identifiability, Monte Carlo variance in simulation summaries, and introduction dispersion across replicates are three separate concerns; the campaign scale addresses variance in the summaries used for fitting, not the underlying parameter identifiability or the introduction-size distribution assumed by the boundary campaigns. If four interacting parameters are inferred from two aggregate summaries, the local sensitivity matrix has rank at most two. Repeated runs reduce Monte Carlo error around those summaries; they do not create independent information directions. The defensible claim is that external biology and observation rules reduce the effective calibration problem: dose-response shape, shedding timing, and non-susceptibility are constrained from literature, while d_{adj} and dispersion-like introduction behavior are bounded by replicated sweeps and posterior surfaces. The remaining uncertainty should be reported as a compatible region, not as a unique mechanistic estimate [48, 49, 62, 93, 94].

9.5 Limitations and implications

Several limitations constrain the interpretation. Norovirus is the only pathogen calibrated to a cruise-specific, response-shadowed empirical endpoint in this monograph. SARS-CoV-2, influenza, and measles use literature-informed natural history, route assumptions, prevalence anchors, and boundary surfaces, but they have not been calibrated to comparable multi-platform cruise datasets. Diamond Princess and other COVID-19 voyages provide valuable respiratory evidence, yet route attribution remains disputed, with mechanistic analyses supporting substantial aerosol contribution and cabin-distribution analyses arguing against appreciable cross-cabin HVAC spread [18, 20, 21, 23]. Influenza and measles ship reports support importation and amplification risk, but they do not supply the same response-calibrated surface as VSP norovirus data [27–29, 102].

Instrument assumptions are also provisional. The wearable sensitivity and specificity values used in the decision model are scenario parameters, not ship-validated estimates for each pathogen. Published wearable studies support physiological anomaly detection for respiratory infection, but performance depends on adherence, baseline windows, algorithms, symptom physiology, and de-

ployment context [80, 81]. Wastewater assumptions face a different problem: detection is feasible, but conversion from signal to infectious prevalence depends on shedding, dilution, sampling design, assay recovery, and action thresholds [78, 170]. The present campaigns therefore support comparative decision logic more strongly than they support absolute claims about a specific commercial instrument.

The model represents crew as a distinct population with role-specific mobility, dining, accommodation, and response parameters, but it does not yet implement full agent-level front-of-house/back-of-house segmentation with role-specific work-exclusion policies and mandatory sick-leave thresholds. Statements about crew segmentation in this monograph refer to the passenger/crew split and the pathway-level food-handler amplification described in Chapter 2, not to a finer FOH/BOH agent partition. The voyage model omits dynamics that would likely matter in operational deployment. Port calls are represented only as stubs, so shore excursions, tenders, crew liberty, local exposure, and port-health interactions are not yet modeled as introduction or transmission processes. Inter-voyage persistence is not active in the monograph campaigns, although cruise norovirus literature documents consecutive-voyage outbreaks and environmental or crew-mediated continuity [12, 14, 16]. The model also does not yet validate respiratory airborne mass against shipboard environmental sampling, nor does it calibrate passenger and crew care-seeking separately across jurisdictions. These omissions matter most when the question shifts from relative policy ranking to vessel-specific operating guidance.

The synthesis nevertheless changes how cruise-outbreak models should be used. Calibration should target the process that generated the data, not the latent epidemic one wishes had been observed. Mid-voyage surveillance should be evaluated for response quality and decision accuracy, with final attack-rate gains interpreted in light of saturation. Pre-boarding screening should be evaluated as movement along a platform- and pathogen-specific introduction surface, with rollover treated as a signal to change the voyage-level decision rather than to intensify the same screening policy. For future work, the highest-value empirical additions are line lists with timing, cabin and venue exposure, linked environmental or wastewater measurements, denominator-based wearable performance on ships, port-call exposure records, and inter-voyage environmental reset data. Those data would add information directions that aggregate VSP counts cannot supply, and would allow CTB to move from response-shadowed calibration toward mechanism-level validation.

Chapter 10

Future Work

10.1 From closed voyages to open-boundary systems

The next version of CTB should treat a voyage as an open epidemiological system rather than a closed experiment. The present campaigns represent embarkation and disembarkation as boundary conditions and leave port visits as stubs. That abstraction was sufficient for comparing response thresholds and screening portfolios, but it is no longer sufficient for vessel-specific guidance. Future work should add explicit flows at embarkation, port calls, tenders, excursions, crew liberty, medical evacuation, and final disembarkation. Each flow should carry both infection state and observation state: infectious but asymptomatic travelers, recently symptomatic passengers who self-defer, crew returning from shore leave, and environmental material moved through laundry, provisions, luggage, and waste streams. The resulting model should distinguish three introduction routes. First, travelers can board while infectious or presymptomatically shedding. Second, passengers and crew can acquire infection during port calls and return to the ship during the same voyage. Third, infection can leave the vessel and re-enter through a later cohort, a later port, or the same crew network. The cruise epidemiology literature already supports this framing because outbreak size depends on voyage duration, passenger and crew denominators, reporting windows, and ship operations rather than on pathogen biology alone [5, 10, 24, 50].

A boundary-flow module should be calibrated against port-specific prevalence nowcasts and itineraries. For respiratory pathogens, the model should incorporate airport and hotel exposure before boarding, shore-excursion group structure, and local respiratory activity at each port. For norovirus, port calls should also expose agents to food-service venues, small boats, visitor centers, and shared restrooms. A useful first implementation is a discrete event layer: every boundary event samples new exposures, modifies contact schedules, and records an observable audit trail. The audit trail should include port, time ashore, excursion group, dining cohort, crew work group, and whether the agent crossed a symptom-reporting or testing interface. These records would allow model outputs to be compared with prospective line lists instead of only final attack rates.

10.2 Inter-voyage persistence and same-ship recurrence

Inter-voyage persistence is the most direct extension of the boundary model. Historical cruise reports describe norovirus outbreaks recurring across consecutive voyages, consistent with incomplete environmental reset, continuing crew infection, repeated passenger exposure, or a mixture of these processes [12, 14, 16]. The 2026 National Geographic *Sea Bird* sequence provides a recent operational example that should be encoded as a validation target. CDC reported a norovirus outbreak on the May 26–31, 2026 voyage, with 9 of 66 passengers and 3 of 24 crew reporting illness. Because cases continued, VSP conducted a field response covering subsequent May 31–June 5 and June 5–10 voyages. CDC then reported another norovirus outbreak on the June 25–30 voyage, with 19 of 68 passengers and 2 of 27 crew reporting illness [173, 174]. These reports do not prove a single uninterrupted transmission chain, but they show why consecutive-voyage state variables are necessary.

The persistence module should retain crew, cabins, surfaces, equipment, and selected environmental reservoirs across voyages. Passenger agents should turn over, but crew agents should carry infection history, symptom-reporting history, work assignment, immunity, and isolation compliance. Surface and wastewater states should be partially reset according to cleaning logs, terminal disinfection, dry-dock events, linen handling, and elapsed time. Model validation should compare not only whether recurrence occurs, but also whether recurrence concentrates in the same crew departments, decks, cabins, dining rooms, or high-touch surfaces. This design would convert recurrence from a qualitative explanation into a measurable hypothesis: persistent crew infection, persistent environmental contamination, and repeated port introduction should generate different timing and spatial patterns.

10.3 Empirical calibration of environmental surveillance

Environmental surveillance should become a calibration target rather than only an alerting instrument. Wastewater, surface sampling, and air or filter sampling provide observations on pathogen burden before all infected persons have presented to medical staff. For cruise ships, Zafeiriadou and colleagues’ 2026 wastewater study and related passenger-ship wastewater work show the practical feasibility of shipboard viral monitoring, while general wastewater surveillance studies define the interpretation problem: nucleic-acid signal is shaped by shedding, dilution, sampling design, assay recovery, and infection-age distribution [32, 152, 170]. The wastewater detector used in the intervention campaigns is a threshold-based positivity flag rather than a quantitative signal-to-prevalence model; the model should therefore stop treating wastewater as a simple binary detector. It should simulate pathogen shedding into ship-specific plumbing catchments, wastewater holding tanks, sampling times, assay limits of detection, and reporting delays. The same principle applies to surface sampling. A positive swab should be modeled as an imperfect measurement of deposition and cleaning history, not as a direct count of infectious dose.

A practical calibration campaign would pair daily syndromic reports with repeated wastewater and surface samples from bathrooms, buffet areas, crew messes, medical spaces, and selected cabins. The statistical target should be a joint likelihood or synthetic-likelihood objective over clinical counts, wastewater cycle-threshold values, surface positives, and intervention timestamps. This would help separate high shedding from high incidence, early detection from prolonged residual signal, and environmental persistence from repeated introductions. It would also test whether advanced surveillance adds most value through earlier detection, route attribution, or avoidance of unnecessary escalation.

10.4 Crew segmentation and crew-specific transmission

Crew are not a homogeneous compartment. The current passenger–crew split, with pathway-level BOH food-handler amplification but without full agent-level FOH/BOH segmentation, should be extended to explicit front-of-house and back-of-house crew segments with finer work groups where data permit. Chapter 2’s use of BOH/FOH terminology at the pathway level should be read as consistent with this future extension, not as a claim that agent-level segmentation is already implemented. Front-of-house crew interact repeatedly with passengers in dining rooms, bars, house-keeping, entertainment, excursions, and medical care. Back-of-house crew interact through galleys, laundry, waste handling, engine and hotel operations, crew messes, and dense accommodation. Both groups can bridge otherwise separated passenger cohorts, but they do so through different contact schedules, reporting incentives, work-exclusion policies, and isolation costs. Diamond Princess analyses and cruise intervention models show that crew mobility and occupational mixing can sustain transmission after passenger restrictions begin [19, 24]. Norovirus cruise studies also show lower average crew attack rates than passenger attack rates, while repeated-voyage outbreaks indicate that crew continuity can still matter [5, 10, 14].

Future CTB campaigns should represent FOH and BOH agents with separate cabins, dining areas, shifts, symptom-reporting probabilities, medical access, wage or employment constraints, and replacement policies. Food handlers require additional states because exclusion from work changes both transmission and ship operations. Housekeeping staff require route-specific exposure through cabins, linens, bathrooms, and high-touch surfaces. A useful empirical partner dataset would include anonymized rosters, work assignments, cabin assignments, sick-call timestamps, isolation orders, return-to-work dates, and department-level testing or swabbing. The goal is not to stigmatize crew, but to model the operational fact that crew are the persistent population on a vessel whose passenger population turns over.

10.5 Prospective validation with cruise-line partners

The most important empirical advance is direct prospective validation. Retrospective public VSP summaries are invaluable for calibration, but they are threshold-selected, clinic-ascertained, and compressed into voyage-level denominators. A cruise-line partnership should collect data before, during, and after voyages under a prespecified protocol. Minimum data elements include embarkation screening outcomes, daily symptom reports, sick-call logs, cabin and travel-party identifiers, dining assignments, excursion groups, crew work schedules, isolation and cleaning timestamps, diagnostic results, and environmental samples. The study should include below-threshold voyages as well as outbreaks, because fade-outs and false alarms are needed to validate outbreak probability and decision quality.

Prospective validation should use registered endpoints. Primary endpoints should include outbreak occurrence, time to first signal, time to response, final reported attack rate, isolation-days, threshold crossing, and operational cost. Secondary endpoints should include cabin secondary attack rate, crew-department incidence, wastewater lead time, surface-sampling concordance, and route-attribution posterior uncertainty. The statistical design should reserve ships, seasons, or itineraries for external validation rather than tuning every platform on the same data. This is also the setting in which parameter identifiability should be revisited. Aggregate attack rate and outbreak probability cannot separately identify all transmission, susceptibility, and contact parameters; individual-level exposure, timing, and environmental data add information directions that the current summary targets lack [47, 62].

10.6 Multi-pathogen and genotype-specific extensions

The multi-pathogen support described in Chapter 2 (route weights, dose response, and per-pathogen infection state) enables the profile schema needed for co-circulation, but the monograph campaigns use single-pathogen introductions. Future CTB should activate concurrent multi-pathogen introductions and evaluate them across the campaign design rather than only through the architecture. A realistic winter voyage can face simultaneous norovirus, SARS-CoV-2, influenza, and other respiratory infections. Co-circulation changes decision quality because one signal can trigger a response for the wrong pathogen, diagnostic capacity can be consumed by a first syndrome, and isolation rules differ between acute gastroenteritis and respiratory disease. The next model should allow simultaneous seeding of enteric and respiratory pathogens, shared symptom-reporting channels, multiplex testing queues, overlapping isolation resources, and syndrome-specific cleaning or ventilation actions. The economic endpoint should include both missed accelerating outbreaks and unnecessary escalation triggered by a low-risk or wrong-pathogen signal.

Norovirus genotype should also become an explicit model dimension. The biology literature supports separating infection, shedding, symptoms, immunity, and severity rather than using a single genotype-independent transmissibility multiplier [51, 59, 98, 99]. Preston et al. provide the direct

cruise-ship motivation for comparing GII.4 and GII.17, while severity studies support genotype-specific cost and clinical-burden multipliers rather than assuming that all genotypes impose the same operational loss [142, 144]. A genotype module should encode genotype-specific susceptibility, prior immunity, symptom probability, vomiting probability, illness duration, shedding curve, and medical-care cost. The cost model should then distinguish a high-transmission genotype from a high-severity genotype. This distinction matters because an operator’s optimal response may differ when the same reported attack rate implies different hospitalization risk, passenger compensation, cleaning burden, or reputational cost.

10.7 Individual-level validation and transmission reconstruction

Mechanistic validation requires individual-level data. Contact tracing, venue attendance, cabin co-occupancy, diagnostic timing, symptom onset, and genomic epidemiology can test whether simulated infections occur in the right people, places, and sequence. For norovirus, sequencing can separate persistence from repeated introduction and can identify mixed-genotype outbreaks. For respiratory pathogens, genomic distance and onset timing can help separate onboard transmission from co-primary infections introduced at embarkation. Transmission reconstruction also supplies the secondary-case distribution needed to validate overdispersion and superspreading rather than fitting only a mean attack rate [65, 91].

The model should output candidate transmission trees, not only case counts. Each simulated infection should retain plausible source agents, exposure route, venue, dose contribution, and observation pathway. Validation can then compare distributions: zero-transmitter fraction, largest cluster, crew-to-passenger links, passenger-to-crew links, cabin-pair transmission, and setting attribution. This approach avoids forcing all heterogeneity into a fitted dispersion parameter when some of it should emerge from schedules, cabins, dining, symptoms, and environment.

10.8 Dining rotation, food contamination, and economic optimization

Dining remains a high-priority structural uncertainty. The current model should replace fixed dining assignment with an empirically parameterized dining-rotation probability. Passengers may return to the same table, rotate among venues, join excursion companions, or change behavior after illness messaging. Food contamination is already implemented as a separate route in Chapter 2, with venue-specific food pools, back-of-house shedding into those pools, and per-agent ingestion dose in Dining-type zones. Future work should extend the current pathway—utensils, buffet service, shared condiments, and vomit or surface contamination sub-pathways within the existing food route—rather than treat food contamination as a new pathway. The model should estimate how much dining rotation is needed to reproduce observed cabin and venue clustering while maintaining

plausible crew workloads. Cruise norovirus reviews identify person-to-person spread, cabins, food-service settings, behavior, and environmental contamination as linked control targets [10]. Primary cruise reports document multiple plausible routes and settings but should be read as outbreak evidence rather than as reviews of interaction mechanisms [5, 12].

The final future-work target is economic optimization. The present decision model compares a finite set of policies. A deployment model should search the Pareto frontier of surveillance portfolios by platform class, itinerary, pathogen mix, and prevalence state. Candidate controls include symptom questionnaires, incentives for wearable adoption, rapid antigen or multiplex testing, wastewater sampling frequency, surface sampling, medical-staff thresholds, crew testing, dining changes, sanitation surge, ventilation actions, and pre-departure delay. Objectives should include expected cases, severe outcomes, VSP or public-reporting probability, unnecessary escalation, passenger compensation, isolation-days, staff burden, test cost, and voyage cancellation risk. The output should not be a universal best policy. It should be a platform-specific frontier: expedition ships may value early interception of one infectious boarder, while mega ships may need prevalence-contingent voyage-level actions once introduction saturation makes individual screening less effective. This optimization step is where CTB can move from scenario analysis to decision support, provided the empirical validation steps above narrow the uncertainty that currently dominates mechanism-level interpretation.

Appendix A

Appendices

The appendices consolidate the tabular reference material that the main chapters cite but do not reproduce in full. Section A gives the platform configuration tables moved from Chapter 3 in response to reviewer comment. Section B collects the ten pathogen profiles from Chapter 4. Section C enumerates the surveillance strategies used across the calibration and intervention campaigns. Section D summarizes the campaign ledger and the two targeted validation/robustness analyses that together produced the 77,310 runs underlying the monograph. Section E gives the boundary outbreak-surface tables for the four calibrated pathogens on the four platforms. Section F states the two-stage Stan model used to summarize the outbreak surfaces in Chapters 5 and 9.

A.1 Platform configuration tables

Tables A.1–A.4 describe the four cruise platforms used in the monograph campaigns. The platform layer is defined in `data/platforms/` and `voyage_config.json` in the open-source repository. Zone counts, dining venue mix, and medical-response defaults are read verbatim from those files; see Chapter 3 for interpretation.

Table A.1: Nominal population, deck geometry, and cabin-corridor structure for the four cruise platforms used in campaigns C7–C14b and the boundary k -sweep. Passenger and crew counts are the nominal complement assumed by the population initializer; realized counts vary by campaign seed. Cabin-corridor resolution replaces earlier well-mixed berthing blocks.

Platform	Agents	Passengers	Crew	LOA (m)	Beam (m)	Cabin/crew corridors
Expedition (<code>expedition_cruise_450</code>)	450	300	150	160	21	12 + 4
Classic (<code>classic_cruise_1900</code>)	1,910	1,350	560	238	32	36 + 6
Spirit (<code>spirit_cruise_3000</code>)	3,000	2,100	900	290	36	48 + 9
Mega (<code>mega_cruise_5000</code>)	7,000	5,000	2,000	Oasis class		81 + 12

Table A.2: Zone-type inventories per platform, extracted from `spatial_layout.json`. “Cabin_Corridor” aggregates passenger and crew corridor segments. “Free” covers public spaces that are not classified as dining or medical (theatres, lounges, promenades, deck spaces). Isolation-unit capacity is a nominal configuration value used by the isolation logic and is zero by default; campaigns override it through response blocks.

Platform	Total zones	Cabin_Corridor	Dining	Medical	Free/Public
Expedition	33	16	4	1	12
Classic	61	42	6	2	11
Spirit	86	57	10	2	17
Mega	129	93	13	2	21

Table A.3: Dining service composition by meal and platform. MDR, buffet, and specialty columns are passenger venue-allocation weights within each meal. The expedition crew-mess entries are an expedition-specific shared-service allocation in the same configuration table; dashes for other classes mean passenger dining weights are normalized over passenger venues and crew dining is handled by the separate crew schedule rather than by this passenger-weight denominator. Weights are read from `voyage_config.json:dining_meal_weights`; rows sum to 1.0 within rounding for the implemented role scope.

Platform	Meal	MDR	Buffet	Specialty	Crew mess
Expedition	Breakfast	0.85	0.00	0.00	0.15
	Lunch	0.70	0.00	0.00	0.30
	Dinner	0.90	0.00	0.05	0.05
Classic	Breakfast	0.20	0.75	0.05	–
	Lunch	0.25	0.60	0.15	–
	Dinner	0.60	0.20	0.20	–
Spirit	Breakfast	0.18	0.80	0.02	–
	Lunch	0.25	0.65	0.10	–
	Dinner	0.65	0.25	0.10	–
Mega	Breakfast	0.15	0.70	0.15	–
	Lunch	0.20	0.50	0.30	–
	Dinner	0.45	0.20	0.35	–

Table A.4: Per-platform medical-response defaults from `medical_response_spec`. Values are the class gradient used when a campaign does not override the block. `sick_call_probability` is per epoch (1 h). `detection_delay_epochs` is a symptom-onset-to-sick-call floor. `crew_screening_interval_epochs` was null in all monograph campaigns. These parameters enter the syndromic surveillance path; the VSP threshold acts independently.

Platform	Sick-call prob.	Isolation compliance	Detection delay (epochs)	Crew screening
Expedition	0.90	0.95	1	off
Classic	0.70	0.90	2	off
Spirit	0.50	0.85	2	off
Mega	0.40	0.80	3	off

A.2 Pathogen profile summary table

Table A.5 lists the ten pathogen profiles in `data/pathogens/edison_10pathogen_profiles.json`. Dose-response families are those in Chapter 4 (Eqs. 4.1–4.2). Route weights sum to 1.0 across direct contact (D), droplet (Dr), HVAC airborne (H), fomite (F), food contamination (Fo), and environmental reservoir (E). The table reports the current profile-schema values; where prose interprets a stress-test profile more narrowly (for example Ebola as body-fluid/contact-dominant), small nonzero schema weights are treated as placeholders for sensitivity analysis rather than calibrated routine routes. `dose_adj` is the configuration value shipped with the profile; the calibrated norovirus value used in the monograph is $d_{\text{adj}} = 10.6$, not the profile default. Non-susceptibility is the innate fraction not participating in transmission; time-varying immunity (vaccination, prior infection) is applied separately by campaign.

Table A.5: Pathogen profile summary. Dose-response parameters are (α, β) for beta-Poisson and single-rate k for exponential. Weights are given as direct/droplet/HVAC/fomite/food/env.

Pathogen	Category	Dose resp.	Parameters	d_{adj}	Non-susc.	Route weights (D/Dr/H/F/Fo/E)
Norovirus GII.4	Enteric viral	β -Poisson	$\alpha=0.111, \beta=32.81$	4.0/10.6	0.20 (FUT2)	0.35/0.10/0.05/0.30/0.20/0.00
SARS-CoV-2	Respiratory viral	β -Poisson	$\alpha=0.18, \beta=58.0$	3.0	0.00 (t-var.)	0.25/0.30/0.30/0.10/0.00/0.05
Influenza A	Respiratory viral	exponential	$k=0.18$	1.5	0.00 (strain)	0.20/0.35/0.30/0.15/0.00/0.00
Measles virus	Respiratory viral	exponential	$k=0.5$	2.0	0.00 (MMR)	0.10/0.35/0.50/0.05/0.00/0.00
<i>L. pneumophila</i>	Bact. water-borne	exponential	$k=0.059$	1.0	0.00	0.00/0.00/0.00/0.00/0.00/1.00
<i>Vibrio</i> spp.	Enteric bacterial	β -Poisson	$\alpha=0.25, \beta=16.2$	1.0	0.00	0.10/0.00/0.00/0.15/0.70/0.05
<i>C. jejuni</i>	Enteric bacterial	β -Poisson	$\alpha=0.145, \beta=7.59$	1.0	0.00	0.15/0.00/0.00/0.20/0.60/0.05
<i>C. difficile</i>	Enteric bacterial	exponential	$k=0.001$	1.0	0.00	0.25/0.00/0.00/0.45/0.00/0.30
Andes hantavirus	Respiratory viral	exponential	$k=0.01$	1.0	0.00	0.15/0.25/0.45/0.10/0.00/0.05
Ebola virus	Filovirus	exponential	$k=0.5$	1.0	0.00	0.55/0.20/0.05/0.20/0.00/0.00

Calibration anchors and cruise-specific attack-rate targets are given in Chapter 4, Table 4.1. The six non-boundary pathogens do not have numeric cruise attack-rate targets in the supplied calibration file; their profiles are retained as structured priors and stress-test cases, per Chapter 4.

A.3 Surveillance strategy definitions

Twenty-three surveillance strategy labels appear across the calibration and intervention campaigns. Table A.6 groups them into three families: uncontrolled baselines, syndromic variants used to probe the observation process (sick-call probability, detection delay, isolation compliance, and VSP threshold), and cascade families that layer diagnostic testing, wearable adoption modes, and wastewater sequencing on top of the syndromic backbone. The syndromic variants sweep one parameter at

a time while holding the others at the platform-specific defaults in Table A.4; the numeric suffix encodes the swept value.

Table A.6: Surveillance strategies. Suffixes: **c{n}** isolation compliance = $0.\{n\}$; **d{n}** detection delay = n epochs; **p{n}** sick-call probability = $0.\{n\}$; **vsp{n}** VSP trigger threshold = $n\%$; **vsp_off** disables the VSP backstop. Cascade variants share the four-tier diagnostic cascade (wearable $\rightarrow$ NP/RDT $\rightarrow$ confirmatory $\rightarrow$ full SOP) and differ in the sampling or biosensor layer.

Strategy	Family	Description
none_true	Baseline	True uncontrolled baseline. No sick call, no isolation, no VSP threshold. Used only to expose latent transmission for calibration and boundary surfaces.
syndromic	Syndromic	Platform-default syndromic surveillance: sick call, isolation compliance, and detection delay from Table A.4, with the VSP 3% backstop active.
synd_stock	Syndromic	Legacy stock defaults: sick-call 0.5, isolation 0.85, detection delay 2 epochs; retained for continuity with pre-spec calibrations.
synd_c50/c70/c95	Syndromic sweep	Isolation compliance held at 0.50, 0.70, or 0.95 while other parameters take platform defaults.
synd_d0/d2/d4/d8/d12	Syndromic sweep	Detection delay held at 0, 2, 4, 8, or 12 epochs.
synd_p30/p50/p90	Syndromic sweep	Sick-call probability held at 0.30, 0.50, or 0.90.
synd_vsp1	Syndromic sweep	VSP threshold lowered to 1%.
synd_vsp5	Syndromic sweep	VSP threshold raised to 5%.
synd_vsp_off	Syndromic sweep	VSP threshold disabled while retaining sick call and isolation.
cascade	Cascade	Four-tier diagnostic cascade on top of syndromic response; clinical rapid diagnostic followed by confirmatory testing and full SOP.
cascade_mpx	Cascade	Multiplex-panel variant; faster pathogen identification at the confirmatory step.
cascade_wear_byod	Cascade + wearable	Bring-your-own-device wearable adoption; anomalies feed the cascade at the wearable tier.
cascade_wear_incent	Cascade + wearable	Incentivized wearable adoption ($\geq 70\%$ passenger uptake) feeding the cascade.
cascade_wear_crew	Cascade + wearable	Crew-mandatory wearable adoption with the same cascade feed.
cascade_ww	Cascade + env.	Wastewater sequencing surveillance layered on the cascade; environmental positivity triggers escalation.

Interpretation and results for these strategies appear in Chapter 7. The syndromic sweeps in particular are the source of the Chapter 5 finding that medical-response knobs are absorbed by the VSP threshold once the backstop fires.

A.4 Campaign manifest summary

Table A.7 gives the run-count ledger for the simulation campaigns whose combined output underlies the monograph. Total run count is 77,310, matching the figure cited in Chapters 1, 2, and 9. The boundary k -sweep is one logical campaign but is split into four pathogen rows because the Stage A Stan summaries use pathogen-specific replicate counts.

Provenance for each campaign is retained in the corresponding **c{n}_campaign_results.csv**, **trek_campaign_results.csv**, **user_uploads/analysis/**, and **vsp_degradation/analysis/** bundles. Note that the four boundary k -sweep rows are sometimes reported as a single “17,600-run boundary campaign” (Chapter 7, Section “Outbreak surfaces from the k -sweep campaign”); the

Table A.7: Campaign manifest run-count ledger. C-numbered campaigns follow the CTB internal chronology; the boundary k -sweep is one logical campaign reported by pathogen to preserve the Stan replicate counts.

Campaign	Purpose	Runs
C7 (3-platform recalibration)	Post-density-model recalibration across expedition, classic, and mega.	5,250
C8 (CONTAM airborne validation)	Native vs. CONTAM analytical mass balance on airborne pathogens.	2,295
C9 (dining and route weights)	Fine-grained sensitivity across dining assignments and route weights.	8,745
C10 (introduction and immunity)	Introduction size and pre-existing immunity sweep.	1,140
C12 (full recalibration)	Seven-pathogen d_{adj} recalibration after ship-operations spec.	6,720
C12b (syndromic-only)	Syndromic-only recalibration under platform-default response.	3,900
C12c (fine norovirus)	Fine-grained norovirus calibration; converged on $d_{\text{adj}} = 10.6$.	5,920
Boundary norovirus k -sweep,	$P_{\text{trigger}}(k)$ and $E[A(k)]$ under VSP baseline response.	2,400
Boundary SARS-CoV-2 k -sweep,	Boundary surface under VSP baseline response.	4,800
Boundary influenza k -sweep, in-	Boundary surface under VSP baseline response.	4,800
Boundary measles k -sweep,	Boundary surface under VSP baseline response with 92% non-susceptibility.	5,600
C14 (mid-voyage intervention)	Detection and intervention sweep across pathogens, platforms, and surveillance strategies.	8,440
C14b (intervention add-on)	Extension of C14 covering additional strategies and immunity levels.	4,220
Starfleet stress-test	Constitution- and Galaxy-class generalization (Chapter 8).	5,520
Synthetic recovery	Six-vector identifiability recovery diagnostic (Chapter 5).	1,200
VSP degradation sweep	Factor-at-a-time and combined VSP robustness panels (Chapter 5).	6,360
Total		77,310

disaggregation here preserves the pathogen-level replicate counts used for the Stan summaries.

A.5 Outbreak surface data

Tables A.8–A.9 give the full $P_{\text{trigger}}(k)$ and $E[A(k)]$ surfaces used in Chapter 5, Table 5.1, and in the decision analysis of Chapter 7. Values are means across replicated runs at each $(k, \text{platform})$ cell under the VSP baseline response.

Table A.8: $P_{\text{trigger}}(k)$: probability of VSP threshold response at introduction size k under baseline VSP settings. Cells report mean triggering probability across replicated seeds. The measles k grid extends further because introductions must clear the 92% non-susceptibility floor. Missing $k = 6$ and $k = 8$ – $k = 9$ cells reflect the sparse k grid used for these surfaces.

Pathogen / platform	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 7$	$k = 10$	$k = 15$
Norovirus, expedition	0.46	0.73	0.87	0.93	0.95	1.00	–	–
Norovirus, classic	0.49	0.79	0.89	0.95	0.98	1.00	–	–
Norovirus, spirit	0.49	0.73	0.84	0.89	0.91	0.99	–	–
Norovirus, mega	0.48	0.71	0.84	0.91	0.94	0.99	–	–
SARS-CoV-2, expedition	0.03	0.19	0.63	0.96	0.99	1.00	–	–
SARS-CoV-2, classic	0.05	0.12	0.16	0.15	0.24	0.31	–	–
SARS-CoV-2, spirit	0.07	0.11	0.13	0.21	0.29	0.34	–	–
SARS-CoV-2, mega	0.08	0.10	0.21	0.22	0.26	0.37	–	–
Influenza, expedition	0.79	0.98	1.00	–	1.00	1.00	1.00	–
Influenza, classic	0.85	0.97	1.00	–	1.00	1.00	1.00	–
Influenza, spirit	0.90	0.98	1.00	–	1.00	1.00	1.00	–
Influenza, mega	0.87	0.99	1.00	–	1.00	1.00	1.00	–
Measles, expedition	0.16	0.35	0.73	–	1.00	1.00	1.00	1.00
Measles, classic	0.17	0.28	0.42	–	0.68	0.71	0.88	0.99
Measles, spirit	0.17	0.39	0.40	–	0.63	0.80	0.85	0.95
Measles, mega	0.16	0.27	0.46	–	0.64	0.81	0.92	1.00

Per-cell replicate counts are 200 runs for SARS-CoV-2, influenza, and measles and 100 runs for norovirus (Chapter 5). The acceleration probability $P_{\text{accel}}(k)$ and expected cost $E[C(k)]$ are stored alongside these tables in `outbreak_surface.csv` but are not reproduced here; the endpoints at k_{max} are already summarized in Table 5.1.

Table A.9: $E[A(k)]$: expected final attack rate (proportion) at introduction size k under the VSP baseline response. Influenza saturates near $A = 0.65$ once triggered; norovirus reaches 0.15–0.18 at $k = 7$; SARS-CoV-2 has the lowest expected final attack rate on expedition and higher values on large ships, while measles remains suppressed by high non-susceptibility.

Pathogen / platform	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 7$	$k = 10$	$k = 15$
Norovirus, expedition	0.054	0.099	0.119	0.140	0.160	0.176	–	–
Norovirus, classic	0.051	0.095	0.120	0.143	0.157	0.179	–	–
Norovirus, spirit	0.036	0.063	0.084	0.098	0.107	0.146	–	–
Norovirus, mega	0.039	0.066	0.085	0.112	0.115	0.145	–	–
SARS-CoV-2, expedition	0.011	0.034	0.043	0.052	0.059	0.079	–	–
SARS-CoV-2, classic	0.026	0.063	0.094	0.093	0.140	0.166	–	–
SARS-CoV-2, spirit	0.041	0.068	0.084	0.137	0.173	0.204	–	–
SARS-CoV-2, mega	0.056	0.076	0.147	0.160	0.184	0.259	–	–
Influenza, expedition	0.512	0.638	0.648	–	0.651	0.651	0.651	–
Influenza, classic	0.550	0.628	0.650	–	0.650	0.650	0.650	–
Influenza, spirit	0.582	0.634	0.650	–	0.650	0.650	0.650	–
Influenza, mega	0.566	0.644	0.650	–	0.650	0.650	0.650	–
Measles, expedition	0.011	0.018	0.025	–	0.032	0.040	0.049	0.061
Measles, classic	0.014	0.023	0.034	–	0.055	0.057	0.070	0.077
Measles, spirit	0.014	0.031	0.032	–	0.051	0.064	0.068	0.076
Measles, mega	0.013	0.022	0.037	–	0.051	0.065	0.073	0.080

A.6 Stan model specification

The boundary Stan analyses cited in Chapters 5 and 9 use a two-stage hurdle structure. Stage A is a Bernoulli-logit model for the probability that a voyage produces a triggered outbreak. Stage B is a Beta-regression model for the conditional final attack rate, fit only on outbreak-positive runs. Neither stage calls the agent-based model; both consume campaign summaries produced by the analysis bundle (Appendix D of the CTB Stan spec).

A.6.1 Stage A: Bernoulli-logit outbreak occurrence

For run r let $y_r^{(A)} \in \{0, 1\}$ indicate whether the VSP threshold response fired within the voyage. Then

$$y_r^{(A)} \sim \text{Bernoulli}(\pi_r), \quad (\text{A.1})$$

$$\text{logit}(\pi_r) = \alpha_{\text{platform}[r]}^{(A)} + \beta_k^{(A)} \log(k_r + 1), \quad (\text{A.2})$$

where k_r is the introduction size and $\alpha_{\text{platform}[r]}^{(A)}$ is a platform intercept. The boundary-surface fits reported here hold surveillance at the VSP baseline and use one pathogen-specific immunity setting, so surveillance and immunity effects are not estimated in these separate pathogen fits. The Stage A model intentionally omits a platform $\times$ introduction-size interaction; this keeps the fitted surface stable with the available cell counts but can mask heterogeneous introduction sensitivity across vessel classes. Platform-specific k slopes should be added when larger boundary sweeps are available.

A.6.2 Stage B: Beta regression on conditional attack rate

For runs with $y_r^{(A)} = 1$, let $A_r \in (0, 1)$ be the final attack rate. Then

$$A_r \sim \text{Beta}(\mu_r \phi, (1 - \mu_r) \phi), \quad (\text{A.3})$$

$$\text{logit}(\mu_r) = \alpha_{\text{platform}[r]}^{(B)} + \beta_k^{(B)} \log(k_r + 1), \quad (\text{A.4})$$

with a single dispersion $\phi > 0$. Attack rates A_r are shrunk from $\{0, 1\}$ by the standard $A_r \leftarrow (A_r(n_r - 1) + 0.5)/n_r$ transform before fitting to keep the likelihood well-defined.

A.6.3 Priors and hyperparameters

$$\alpha_p^{(A)}, \alpha_p^{(B)} \sim \text{Normal}(0, 2), \quad (\text{A.5})$$

$$\beta_k^{(A)}, \beta_k^{(B)} \sim \text{Normal}(0, 1), \quad (\text{A.6})$$

$$\phi \sim \text{Exponential}(1/10). \quad (\text{A.7})$$

The Beta-regression precision prior was widened to Exponential(1/10) (prior mean $\phi = 10$) after the initial Exponential(1) prior was flagged as too tight relative to the empirical attack-rate spread; sensitivity to $\phi \in \text{Exponential}(1/5)$ – $\text{Exponential}(1/20)$ did not change platform rankings. Other priors are weakly informative. Because surveillance and immunity are fixed within each reported boundary fit, no γ or δ coefficients are reported for these fits.

A.6.4 Diagnostics and reporting

All Stan fits used 4 chains with 1000 warmup and 1000 sampling iterations per chain. Chains are sampled with NUTS at default `adapt_delta`= 0.95. Convergence is monitored via $\hat{R} \leq 1.01$ and bulk effective sample size ≥ 400 on all reported quantities. The Stage B trajectory models had zero divergent transitions in the available fit summaries. No `fit_status.json` file was present in the revision workspace for Stage A outbreak models, so Stage A divergent-transition counts could not be verified from that file; the convergence criteria above are the reporting thresholds used for the retained summaries. Reported Stage A summaries in Chapter 5 are posterior means of π_r marginalized over k at the observed campaign design points, aggregated by platform. Stage B summaries are posterior means of μ_r under the same design points. Posterior predictive checks compare simulated Stage A outbreak counts and Stage B attack-rate distributions against the empirical campaign summaries; platform-specific posterior predictive intervals are used for the interpretation in Chapter 5, Section “Multi-pathogen boundary surfaces.”

Full run manifests and Stan input schemas are stored under `picard_framework/analysis/stan/` and `user_uploads/stan/`. The four calibrated pathogens (norovirus, SARS-CoV-2, influenza, measles) each have their own Stage A and Stage B fits using the surfaces in Appendix A.5.

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